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HY Endocrine, by Dr Michael D Mehlman

- Hypothalamic hormones and their stimulatory/inhibitory effect on anterior pituitary hormones.			
Hypothalamic hormone		Anterior pituitary hormone	End-organ
Gonadotropic-releasing hormone (GnRH)	(+)	Luteinizing hormone (LH)	- Ovarian theca interna cells + testicular Leydig cells to make androgens.
Gonadotropic-releasing hormone (GnRH)	(+)	Follicle-stimulating hormone (FSH)	- Ovarian granulosa cells + testicular Sertoli cells to make aromatase + inhibin. - Follicular development in females.
Thyrotropin-releasing hormone (TRH)	(+)	Thyroid-stimulating hormone (TSH)	- Thyroid gland to make thyroid hormone.
Growth hormone-releasing hormone (GHRH)	(+)	Growth hormone (GH)	- Causes liver to produce insulin-like growth factor-1 (IGF-1), which causes tissue growth.
Corticotropin-releasing hormone (CRH)	(+)	Adrenocorticotropic hormone (ACTH)	- Causes zona fasciculata of adrenal cortex to make cortisol.
Dopamine	(-)	Prolactin	- Stimulates milk production in breast tissue. - Dopamine inhibits prolactin secretion. - If the pituitary stalk is severed (i.e., we lose signaling from the hypothalamus), prolactin is the only hormone that goes ↑. This is because we lose inhibitory signaling, whereas with the other anterior pituitary hormones, they lose positive signaling.

- Hypothalamic hormones that are stored in the posterior pituitary. - Many students erroneously think vasopressin and oxytocin are <i>produced</i> in the posterior pituitary. - The supraoptic and paraventricular nuclei in the hypothalamus produce both hormones.		
Hypothalamic hormone	Storage location	End-organ
Vasopressin (Anti-diuretic hormone; ADH)	Posterior pituitary	- Inserts aquaporins in medullary collecting duct of kidney, causing ↑ free water reabsorption.
Oxytocin	Posterior pituitary	- Lactation (milk letdown/secretion). - Uterine contractions during labor. - Maternal/infant bonding during breastfeeding.

Hypothalamic-pituitary-gonadal axis	
GnRH	- Gonadotropin-releasing hormone; produced by the hypothalamus. - Stimulates LH and FSH secretion from anterior pituitary. - "Gonadotropins" refer to the peptide hormones, LH and FSH. - Mechanism for amenorrhea in anorexia is ↓ GnRH pulsation. This is because leptin is required for adequate GnRH pulsation, and in anorexia, leptin levels are low. This is because leptin makes you feel full, and in anorexia, the patient is "hungry." ↓ GnRH pulsation causes LH and FSH to both be ↓, which leads to reduced ovarian hormone production and follicle stimulation. - In polycystic ovarian syndrome (PCOS), insulin resistance in high-BMI female causes abnormal GnRH pulsation, which causes ↑ LH/FSH ratio. I discuss this in extensive detail in the HY Repro/Obgyn PDF. - Leuprolide, goserelin, and nafarelin are all GnRH receptor agonists. When administered continuously, they cause desensitization of the GnRH receptor at the anterior pituitary, leading to ↓ LH and FSH secretion (i.e., even though the drugs are pharmacologic agonists, they function clinically as antagonists).

	- GnRH secretion is inhibited by estrogen, androgen, and progesterone. This inhibition is called negative-feedback. - Exogenous anabolic steroids, for instance, $\downarrow$ GnRH secretion $\rightarrow$ $\downarrow$ LH + FSH $\rightarrow$ $\downarrow$ endogenous testicular production of androgens. - If we have central hypogonadism, we will have $\downarrow$ GnRH $\rightarrow$ $\downarrow$ LH + $\downarrow$ FSH $\rightarrow$ $\downarrow$ androgens + $\downarrow$ estrogens. - "Central" means the problem is up at the hypothalamus causing $\downarrow$ GnRH. - If we have primary hypogonadism, we will have $\downarrow$ ovarian/testicular androgens and estrogens leading to $\uparrow$ GnRH $\rightarrow$ $\uparrow$ LH + $\uparrow$ FSH. - "Primary" means the end-organ itself (i.e., ovary/testis) is the cause of the hormonal issue. - Menopause (which is physiologic), as well as any condition characterized by primary hypogonadism or insufficiency (i.e., Turner, Klinefelter, premature ovarian failure, etc., which are all pathologic), will cause $\uparrow$ GnRH secretion due $\downarrow$ negative-feedback. - Androgen-insensitivity syndrome causes $\uparrow$ GnRH due to inability of androgen to exert negative-feedback at androgen receptors at the hypothalamus.
LH	- Luteinizing hormone; produced by anterior pituitary. - Stimulates testicular Leydig cells in males, and ovarian theca interna cells in females, to make androgens. - LH is $\uparrow$ in PCOS, which is why these patients have hirsutism. In PCOS Qs for USMLE, select $\uparrow$ LH and $\downarrow$ FSH. This is due to "abnormal GnRH pulsation." - LH is $\downarrow$ in amenorrhea due to anorexia. In Qs on USMLE, select $\downarrow$ for both LH and FSH. This is due to "decreased GnRH pulsation."
FSH	- Follicle-stimulating hormone; produced by anterior pituitary. - In males, stimulates testicular Sertoli cells to make aromatase and inhibin B; also facilitates sperm maturation during spermatogenesis. - In females, stimulates follicular development and ovarian granulosa cells to make aromatase and inhibin B. - FSH is $\downarrow$ in PCOS for USMLE. The LH/FSH ratio is $\uparrow$. Some students will ask, "well aren't both LH and FSH $\uparrow$, just LH $\uparrow$ more than FSH?" Most literature sources say FSH is straight-up $\downarrow$ in PCOS. I've never seen an NBME Q assess an $\uparrow$ for it. Point is, if I were to sit the real deal again, I'd put a $\downarrow$ for FSH, and if it's wrong, everyone can go fuck themselves.
Androgens	- Produced by testicular Leydig cells in males and ovarian theca interna cells in females. - DHEA-S is an adrenal-specific androgen made in the zona reticularis of the adrenal cortex. The adrenal gland can also make androstenedione. These two androgens can then go peripherally to be converted into testosterone. - In other words, if a patient has hirsutism and the Q tells you DHEA-S is high, you know it is the adrenal gland that is hyper-secreting androgens. - Testosterone is converted into dihydrotestosterone (DHT) via 5-alpha reductase. It is DHT that is the strongest androgen and has the greatest effect at tissues. This enzyme is inhibited by finasteride (used for BPH and androgenetic alopecia). - DHT is necessary for external male secondary sex characteristics, so if DHT is low, then the male will appear phenotypically female. - In 5-alpha reductase deficiency, the karyotypic male will appear female for the first 12 years of life, followed by "penis (phallus) at age 12," where the surge in testosterone from puberty can override the 5-alpha reductase deficiency where a threshold DHT level is achieved, leading to a 3-4-cm clitoral hood. The vignette will tell you a 12-year-old girl has grown 4 inches in past 4 months + has acne + hair on upper lip + clitoral hood growth. Karyotype is 46XY since this is a male. - Androgen-insensitivity syndrome is a karyotypic male (46XY) presenting as phenotypic female due to failure of testosterone and DHT to have effect. Both are $\uparrow$ in AIS due to $\downarrow$ negative-feedback at hypothalamus and anterior pituitary. So LH is also $\uparrow$. I talk about the sex disorders later in this PDF, but AIS will be a 15-year-old girl who's never had a menstrual period + vagina ends in blind pouch + has scanty/absent pubic/axillary hair. - Androgens build muscle, stimulate prostate growth, maintain spermatogenesis, and enhance libido.

	- High androgens (anabolic steroids) accelerate atherosclerotic plaque maturity ($\uparrow$ LDL and TGAs, $\downarrow$ HDL), increase plaque calcium scores (marker of plaque maturation), increase hematocrit (sometimes leading to hyperviscosity syndrome), and can cause liver damage.
Aromatase	- Produced by Sertoli cells in males and granulosa cells in females in response to FSH. - Converts androgens into estrogens. - Testosterone $\rightarrow$ estradiol (potent estrogen). - Androstenedione $\rightarrow$ estrone (less potent estrogen). - DHEA-S $\rightarrow$ estriol (pregnancy-associated estrogen; weakest estrogen). - Anastrozole and exemestane are aromatase inhibitors. - There's an NBME Q that gives gynecomastia in a male where the answer is Sertoli-Leydig cell tumor (granulosa cell tumor not listed). SL tumors normally make androgens. The reason gynecomastia is possible is because of aromatization of androgens into estrogens. - Physiologic gynecomastia can occur in adolescent males, and virginal breast hypertrophy in adolescent females, due to $\uparrow$ androgen production $\rightarrow$ aromatization $\rightarrow$ $\uparrow$ estrogens. - Closure of growth plates occurs via estrogens, via aromatization from androgens.
Estrogens	- Estradiol is produced mainly by the ovaries (and testes in males). - Estrone is produced mainly by adipose tissue. In post-menopausal women, adipose tissue becomes important source of estrogen, where low BMI $\uparrow$ risk of osteoporosis. Interestingly, the NBMEs assess BMI of 20 and lower in young females as $\uparrow$ risk of osteoporosis, even though low BMI is <18.5. - Estriol is produced primarily by the placenta; $\downarrow$ in Down syndrome. - Adrenal DHEA-S (an adrenal-specific androgen) can be aromatized into estradiol. - Estradiol stimulates the endometrial lining to grow during the follicular phase of the menstrual cycle. - Its levels gradually increase during the follicular phase. Once a critical threshold is reached (called the estrogen threshold), its effects at the hypothalamus flip from negative-feedback to positive-feedback, causing the LH surge, which triggers the rupturing of the Graafian follicle. - Stimulates secondary sex characteristics in females – i.e., breast development and changes in body fat distribution. - Maintains bone density by $\uparrow$ OPG/RANK ratio. In post-menopausal women, $\downarrow$ estrogen $\rightarrow$ $\uparrow$ RANK in comparison to OPG $\rightarrow$ $\uparrow$ osteoclast activity $\rightarrow$ osteoporosis. - Normally, RANK-L on surface of the osteoblast binds to RANK receptor on the osteoclast to induce bone resorption. OPG is a RANK receptor decoy that binds RANK-L, thereby decreasing RANK-L-RANK interaction and bone resorption. Estrogen is protective of osteoporosis because it increases OPG. - Estrogen $\uparrow$ HDL and $\downarrow$ LDL. Post-menopausal lipids ($\downarrow$ estrogen) are worse for this reason. - Hormone-replacement therapy (HRT) is only approved for severe perimenopausal vasomotor symptoms (i.e., hot flashes, urge incontinence, atrophic vaginitis). It is not given to help preserve bone density or for positive role on mood and neurocognition, since $\uparrow$ absolute estrogen exposure in women $\uparrow$ risk of breast cancer, MI, and thromboembolic events (i.e., DVT, PE, stroke). The latter is because estrogen upregulates fibrinogen and factors V and VIII. - Unopposed estrogen is the mechanism for endometrial hyperplasia and adenocarcinoma. The three groups this refers to on NBME exams, as per my observation, are 1) PCOS (failure of ovulation leads to absent progesterone), 2) HRT patients who stop taking the progesterone component the way they're supposed to, and 3) granulosa cell tumors (which secrete estrogen). The reason is because estrogen stimulates the growth of endometrium; progesterone inhibits growth of endometrium; so the balance between these two hormones keeps the endometrial lining in check. USMLE might give a vignette of vaginal bleeding in a woman with unopposed estrogen, and the answer will just be "endometrial biopsy" to rule out cancer. - Estrogen-containing OCPs $\downarrow$ protein content of breast milk (asked on 2CK Obygn form), where the risk needs to be discussed with breastfeeding women who want to take OCPs.
Progesterone	- Produced by the corpus luteum (ruptured ovarian follicle) following ovulation and maintains the endometrial lining during the luteal phase of the menstrual cycle. As

	discussed earlier, if fertilization occurs, hCG will maintain the corpus luteum so that progesterone production continues + can maintain the pregnancy. - Also produced in small amounts in the adrenal glands. - Along with estrogen, facilitates proper mammary gland development. During pregnancy, progesterone helps maintain the structure of the mammary glands. - Progesterone inhibits the effects of prolactin at the alveolar cells in the breast (milk-producing cells), preventing lactation during pregnancy. Once the placenta is expelled at parturition and progesterone levels fall, prolactin can induce lactation. - Progesterone-only contraceptives are considered safe while breastfeeding, since the low amount of progesterone does not inhibit lactation, even though the high progesterone during pregnancy does. It is the effect of estrogen in combined OCPs that decreases protein content of breast milk. - Progesterone plays a role in thermoregulation, where $\uparrow$ secretion at ovulation $\uparrow$ body temperature, which is one way to predict timing of ovulation, albeit dubiously. There is an NBME question floating around where they show a graph of increased body temperature at day 14 of menstrual cycle, and answer is just progesterone. Not hard.
hCG	- Human chorionic gonadotropin; produced by syncytiotrophoblastic cells of the placenta. - Maintains the corpus luteum (remnant of ruptured ovarian follicle) after fertilization has occurred. - Corpus luteum normally secretes progesterone, which maintains the endometrial lining during pregnancy. - Without fertilization, there is no placenta, hence no hCG present to maintain the corpus luteum $\rightarrow$ corpus luteum degrades $\rightarrow$ $\downarrow$ progesterone $\rightarrow$ endometrial lining cannot be maintained $\rightarrow$ sloughs $\rightarrow$ menstruation. - hCG peaks at 8-10 weeks of pregnancy. After this point, the placenta takes over production of progesterone. This obviates the need for the corpus luteum. Since we don't need the corpus luteum anymore to make progesterone, we don't need hCG anymore, which is why hCG falls after 8-10 weeks. - High hCG at 8-10 weeks of pregnancy is responsible for nausea and vomiting in pregnancy. If the vomiting is severe enough where the female is ketogenic (i.e., ketones detected in the urine), we call this hyperemesis gravidarum. I will discuss this more later. - Beta-hCG simply refers to the subunit of hCG that is measured during a pregnancy test. This is because hCG shares the same alpha subunit with TSH, LH, and FSH, so we must measure for the beta-subunit to determine whether the hormone is present or not. - An NBME Q gives gynecomastia in a male patient taking hCG. They ask for the mechanism $\rightarrow$ answer = "testis producing estrogen." They can also have as answer "estrogen; direct effect on breast." The reason this makes sense is because, since hCG shares the same alpha-subunit with LH and FSH (and TSH), giving hCG is as though we are giving LH and FSH at the same time. So LH effect $\rightarrow$ androgen production; FSH effect $\rightarrow$ aromatase production $\rightarrow$ therefore androgens are converted to estrogens in the testes $\rightarrow$ mechanism via which we could get gynecomastia. - hCG is $\uparrow$ in Down syndrome during second-trimester quad screen ($\downarrow$ AFP, $\uparrow$ hCG, $\downarrow$ estriol, $\uparrow$ inhibin A). Just remember that the ones that have "Hs" are $\uparrow$ (i.e., hCG and inhibin A both have Hs, so those are the ones that are $\uparrow$).
Inhibin	- Inhibin A and B are produced by Sertoli cells in males and granulosa cells in females. - Both exert negative feedback at the anterior pituitary to specifically $\downarrow$ FSH secretion. - Inhibin A is measured as part of the second trimester quad screen, not inhibin B, because it is more accurate in helping to predict fetal chromosomal abnormalities.
Prolactin	- Produced by anterior pituitary. - Stimulates lactation (milk production). - Levels rise during pregnancy, but lactation is inhibited by $\uparrow$ progesterone causing $\downarrow$ prolactin effects at its receptor on breast alveolar cells. As mentioned above, after expulsion of the placenta and progesterone falls, prolactin can induce lactation. - Dopamine inhibits prolactin. If the pituitary stalk is severed, prolactin $\uparrow$ due to lack of negative-feedback by dopamine.

	- Bromocriptine (D2 receptor agonist) is first-line Tx for prolactinoma (since dopamine inhibits prolactin). - D2 antagonists (i.e., anti-psychotics, metoclopramide) can cause hyperprolactinemia. - Hyperprolactinemia presents as impotence in men, amenorrhea in women, and galactorrhea in both. - Prolactin is ↓ in Sheehan syndrome (discussed later).
Oxytocin	- Produced by hypothalamus and stored in posterior pituitary. - Induces milk secretion (let-down) and uterine contractions. - Suckling reflex at breast causes ↑ oxytocin secretion, causing ↑ milk let-down. - Known colloquially as the "love hormone," since its secretion facilitates maternal-neonate bonding. - Ferguson reflex → cervical distension by the fetus at parturition ↑ oxytocin production, causing ↑ uterine contractions. - Excessive administration during parturition can cause uterine hypertonus and tachysystole, ↑ risk of uterine rupture and fetal complications.

Sheehan syndrome

- Maternal pituitary grows in size during pregnancy due to ↑ hormone production.
- Traumatic labor (e.g., C-section with loss of considerable blood) ↑ risk of anterior pituitary ischemic infarction.
- Textbook vignette is ↓ ability to breastfeed due to ↓ milk production following labor where BP was ↓.
- Schools/resources will usually just teach that prolactin is ↓. But USMLE wants you to know that **all hormones** coming from the anterior pituitary are low.
- NBME wants a ↓ for prolactin, ACTH, and TSH, as well as an ↑ for aldosterone.
- The ↑ for aldosterone is weird, I agree. Take it up with NBME, not me. My suspicion is that ↓ cortisol from ↓ ACTH can lead to ↓ basal blood pressure, where aldosterone could go ↑ to compensate. The patient is not going to have overt hyperaldosteronism with ↑ Na⁺, ↓ K⁺, ↑ HCO₃⁻, and ↑ pH. It's more that, in theory, basal level aldosterone could go ↑ slightly to compensate.
- The vignette can mention fatigue, which reflects ↓ TSH (secondary hypothyroidism).

Acromegaly

- Caused by excess growth hormone secretion (usually by a tumor) following closure of the growth plates (↑ GH secretion prior to growth plate closure causes gigantism).
- As mentioned earlier, GH goes to the liver, which causes it to secrete IGF-1. It is then IGF-1 that induces growth effects at tissues. An NBME Q gives easy vignette of acromegaly and then asks what you check; serum GH is wrong; answer is serum IGF-1.
- Acromegaly causes prognathism (lantern jaw), large ears/nose, arthritis, and carpal tunnel syndrome (due to ↑ joint and tendon growth), enlargement of hands/feet, hypertension, type II diabetes (GH causes insulin resistance), and ↑ risk of arrhythmia and cardiomyopathy.
- Treatment is somatostatin analogue (octreotide). Somatostatin is aka growth hormone-inhibiting hormone (GHIH) and shuts off its secretion from the anterior pituitary. It also ↓ secretion of insulin and glucagon from the pancreas.

Vasopressin conditions

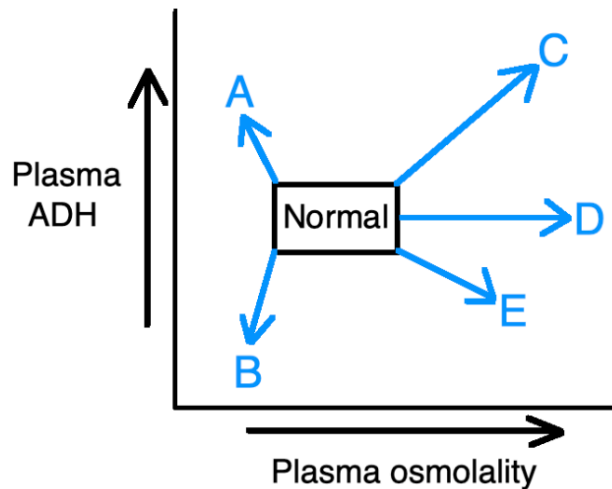
- Vasopressin (aka anti-diuretic hormone; ADH).
- ADH is produced by both the supraoptic nucleus and paraventricular nucleus of the hypothalamus → stored in posterior pituitary.
- ADH ↑ free water reabsorption by the medullary collecting duct (MCD) of the kidney by causing aquaporin insertion.

	- If ADH is $\uparrow$, we get $\uparrow$ free water reabsorption and dilution of our serum $\rightarrow$ serum sodium and osmolality $\downarrow$. Likewise, our urine becomes more concentrated, meaning urine osmolality and specific gravity $\uparrow$. - If ADH is $\downarrow$, we get $\downarrow$ free water reabsorption and concentration of our serum $\rightarrow$ serum sodium and osmolality $\uparrow$. Likewise, our urine becomes more dilute, meaning urine osmolality and specific gravity $\downarrow$. - The primary cells in the brain that recognize serum osmolality and contribute to the secretion of vasopressin are hypothalamic osmoreceptor cells. These are located in the organum vasculosum of the lamina terminalis (OVLT) and the subfornical organ, which detect changes in the osmolality of the blood. $\uparrow$ serum osmolality causes these osmoreceptors to stimulate the nearby supraoptic and paraventricular nuclei.
SIADH	- Syndrome of Inappropriate Anti-Diuretic Hormone secretion $\rightarrow$ means too much ADH (vasopressin secretion). - Central SIADH $\rightarrow$ follows head trauma, meningitis, brain cancer, and pain (latter on 2CK Surg). - Ectopic SIADH $\rightarrow$ small cell lung cancer secreting ADH. - Drug-induced ADH $\rightarrow$ ultra-rare on USMLE, but carbamazepine can do it. - Patient will have dilute serum and concentrated urine: - $\downarrow$ serum sodium, $\downarrow$ serum osmolality, $\downarrow$ serum specific gravity. - $\uparrow$ urinary osmolality, $\uparrow$ urinary specific gravity. - You must know serum sodium is normally 135-145 mEq/L. So in SIADH, it's <135. - Serum vs urinary osmolalities will be all over the place and you do <i>not</i> need to memorize values. They might say serum osmolality is 250 and urinary is 750, and then you say, "Well I can tell serum is dilute compared to urine, which sounds like SIADH." - Specific gravity will be 1.000-1.030 on USMLE. Sounds obscure, but it shows up quite a bit, particularly on 2CK Qs. It's to my observation that values 1.000-1.006ish are "dilute"; 1.024-1.030 are "concentrated." For 9/10 Qs, values will be what you expect. - As I talk about in detail in my HY Arrows PDF, only the medullary collecting duct osmolality will change in response to ADH. The USMLE will ask you for the osmolality of the urine at different nephron locations in comparison to serum, and the combo is: PCT isotonic; juxtaglomerular apparatus (JGA) hypotonic; MCD hypertonic. $\rightarrow$ The PCT is always isotonic no matter what; the JGA (at top of thick ascending limb of loop of Henle) is always hypotonic no matter what; the MCD is clearly hypertonic in SIADH since we're pulling free water out of the urine. - "Fluid restriction" is first answer in diagnosis/management on 2CK. We want to see how serum/urinary values change in response. This will be the answer 9 out of 10 times. - The 1/10 answer is 3% saline (answer on one 2CK NBME), which is when the patient has symptomatic hyponatremia, where there is mental status change (i.e., confusion). I caution you, however, with this answer, since 9 out of 10 times, fluid restriction is correct. Normally, the Q will give no mental status change, or the patient will have overt coma, and they want fluid restriction. But if they talk about confusion in a patient with low sodium, 3% saline is correct on a new 2CK form. It is likely the NBME doesn't want to cavalierly teach hypertonic saline as a correct answer due to the danger of causing central pontine myelinolysis if the hyponatremia is corrected too quickly. - Demeclocycline is answer on NBME for treatment of SIADH. Demeclocycline is technically a tetracycline antibiotic but isn't used because it can cause insensitivity to ADH at the kidney (i.e., nephrogenic diabetes insipidus). So we essentially cause a 2nd problem that cancels out the 1st problem. - Conivaptan and tolvaptan are ADH receptor antagonists that can be used for SIADH. You can be aware they exist, but I haven't seen them assessed on NBME (I'm not talking about Qbank).
DI	- Central diabetes insipidus $\rightarrow$ means not enough ADH secretion by hypothalamus, or the posterior pituitary is unable to release it properly. - Nephrogenic DI $\rightarrow$ insensitivity to ADH at the kidney (serum ADH is $\uparrow$). - Similar to central SIADH, central DI can be caused by head trauma, meningitis, and cancer. - Nephrogenic DI is caused by lithium, demeclocycline, NSAIDs, and hypercalcemia.

	- There is an NBME Q that asks about a patient's response to ADH who is on chronic NSAIDs, and the answer is "$\leftrightarrow$ Response to ADH" and "$\leftrightarrow$ urinary osmolality" (meaning no change). - 2CK Q gives patient with primary hyperparathyroidism and $\uparrow\uparrow$ serum calcium + nephrogenic DI; Q asks cause of the DI $\rightarrow$ answer = hypercalcemia. High calcium can cause renal insensitivity to vasopressin. - Serum vs urinary values are the opposite of SIADH: - $\uparrow$ serum sodium (>145 mEq/L), $\uparrow$ serum osmolality, $\uparrow$ serum specific gravity. - $\downarrow$ urinary osmolality, $\downarrow$ urinary specific gravity. - PCT isotonic; JGA hypotonic; MCD hypotonic. $\rightarrow$ The PCT is always isotonic no matter what; the JGA is always hypotonic no matter what; the MCD is clearly hypotonic in DI since we're not pulling free water out of the urine. - When we are trying to first diagnose DI, the first thing we do is fluid restriction, same as with SIADH. We want to see how serum/urinary values change first. This is the answer 9/10 times. I talk about the exception below. - After we do fluid restriction and determine that the urine is staying dilute + the serum is staying concentrated, the next best step is desmopressin (analogue of vasopressin). If the urine gets more concentrated, (i.e., if the drug works), we know central DI is the diagnosis and we're merely deficient in ADH. - If desmopressin doesn't work, we know we have nephrogenic DI. I should point out that even in nephrogenic DI, desmopressin might work <i>but only very little</i>, whereas with central DI, administration will $\uparrow\uparrow$ urinary osmolality robustly. It will be obvious on USMLE. But my point is, don't say, "Oh well desmopressin worked like 5% so we can't have nephrogenic DI here." - Treatment for central DI is therefore desmopressin. - Treatment for nephrogenic DI is NSAID + a thiazide. Sounds weird, but $\downarrow$ Na^+ reabsorption induced by thiazides in the early-DCT promote compensatory $\uparrow$ Na^+ reabsorption in the PCT, where water follows Na^+ and our net loss of fluid is less than without the thiazide. In healthy individuals, however, they <i>will</i> lose more net fluid with the thiazide. The NSAID presumably $\downarrow$ renal blood flow, which will $\downarrow$ GFR and $\downarrow$ net fluid loss. - There is difficult 2CK NBME Q where they say patient is on lithium + has $\uparrow$ urinary output, and fluid restriction is wrong answer (I say hard because 9/10 times, fluid restriction is correct when it's listed); answer = NSAID + thiazide diuretic. The implication is, if it's obvious the patient's diagnosis is already nephrogenic DI, going straight to Tx is acceptable. This is the 1/10 example.
PP	- Psychogenic polydipsia means the patient is simply drinking too much. - Both the urine and serum will be dilute. - Serum vs urinary values are the opposite of SIADH: - $\downarrow$ serum sodium (<135 mEq/L), $\downarrow$ serum osmolality, $\downarrow$ serum specific gravity. - $\downarrow$ urinary osmolality, $\downarrow$ urinary specific gravity. - I'd say 3/4 Qs on USMLE are obvious and will say some psych patient is drinking lots to "clear himself from evil spirits," etc. - Probably 1/4 Qs won't be an obvious psych vignette, but will just show you the lab values where you have to say, "The urine and serum are <i>both</i> dilute, so this is psychogenic polydipsia." - First step in diagnosis is fluid restriction in order to see how urinary/serum values change.

Mannitol effects on vasopressin

- What USMLE will do is say a patient is given mannitol + then ask you for Δ in serum ADH and osmolality.



- Answer = $\uparrow$ ADH, $\uparrow$ serum osmolality.

- Mannitol is a sugar alcohol. Infusion will cause immediate increase in serum osmolality since we are increasing the solute concentration of the serum (i.e., there's more "stuff" now in the serum).

- An increase in serum osmolality will trigger ADH release, which in turn will reabsorb more free water in the medullary collecting duct of the kidney to help re-dilute the serum back to original osmolality.

- $\uparrow$ serum tonicity + osmolality $\rightarrow$ $\uparrow$ ADH release $\rightarrow$ dilutes serum back to normal.
- $\downarrow$ serum tonicity + osmolality $\rightarrow$ $\downarrow$ ADH release $\rightarrow$ concentrates serum back to normal.

- Mannitol is an osmotic diuretic that acts by exerting oncotic pressure within the PCT and thin descending limb of the loop of Henle, thereby helping to facilitate tubular water retention and diuresis. It is also the answer on 2CK forms for how to reduce intracranial pressure in patients who are already being hyperventilated (lowering CO₂ reduces cerebral perfusion).

- Avoid mannitol in patients with heart failure, since initial increases in plasma volume due to higher serum osmolality will transiently increase preload on the heart.

Thyroid intro points

Blood supply	- Superior part of thyroid: superior thyroid artery (branch of external carotid). - Inferior part of thyroid: inferior thyroid artery (branch of thyrocervical trunk, which is branch of subclavian). - There is an NBME Q floating around somewhere where they ask blood supply.
Embryo/Anatomy	- The thyroid gland originates at the base of the tongue as a small diverticulum. This point of origin at the base of the tongue is called the foramen cecum. - On USMLE, the answer = "endoderm of foramen cecum," when they ask about embryo. - The developing thyroid then descends through the neck, maintaining a connection with the foramen cecum via the thyroglossal duct, before reaching its final position anterior to the trachea and inferior to the hyoid bone, spanning between the C5 and T1 vertebrae. - The relevance for USMLE is that they will give an easy vignette of thyroglossal duct cyst (i.e., painless central neck lump in child that moves upward with swallowing or protrusion of the tongue), and then the answer is "endoderm of foramen cecum."
Physio	- TRH secreted by hypothalamus $\rightarrow$ stimulates TSH secretion by anterior pituitary $\rightarrow$ binds to TSH receptor at thyroid gland $\rightarrow$ activates adenylyl cyclase $\rightarrow$ $\uparrow$ cAMP and protein kinase A (PKA). - PKA stimulates the uptake of iodide ions into the thyroid gland and promotes thyroperoxidase-mediated organification of iodide into I₂.

	- The I₂ is then attached to the tyrosine residues of a large molecule called thyroglobulin. These residues are coupled together to form T₃ (triiodothyronine) and T₄ (thyroxine). - TSH also enhances the proteolysis of thyroglobulin, releasing thyroxine (T₄) and triiodothyronine (T₃) into the bloodstream. - T₃ is stronger than T₄ in terms of physiologic effect but is present in smaller amounts. - Most T₃ and T₄ are protein bound (>99.7%). It is the free hormone that exerts the predominance of physiologic effects. - T₄ can be converted to T₃, but T₃ cannot be converted to T₄. The enzyme via which T₄ cleavage into T₃ occurs is called 5-deiodinase. - Thyroid hormone exerts negative-feedback at the hypothalamus and anterior pituitary, thereby shutting off TRH and TSH, respectively.
Hormone function	- Thyroid hormone is necessary for basal metabolism. During fetal development, it is particularly important for myelin sheath formation and bone growth.

Thyroid endocrine terminology	
Primary hypothyroidism	- ↓ T₃, ↓ T₄, ↑ TSH. - The thyroid gland itself is under-producing thyroid hormone. TSH goes ↑ due to ↓ negative-feedback at hypothalamus + anterior pituitary. - HY examples are Hashimoto thyroiditis (autoimmune destruction of thyroid gland) and iodine deficiency (insufficient thyroid hormone precursor).
Secondary hypothyroidism	- ↓ T₃, ↓ T₄, ↓ TSH. - The thyroid gland is under-producing thyroid hormone due to inadequate stimulation (i.e., the anterior pituitary isn't producing sufficient TSH). - HY causes are Sheehan syndrome (ischemic infarction of anterior pituitary postpartum) and pituitary tumors (e.g., prolactinoma, which impinge on and necrose the thyrotropes in the anterior pituitary that produce TSH). - ↓ TRH from the hypothalamus causing ↓ TSH could be another theoretical cause, but I've never seen this assessed.
Primary hyperthyroidism	- ↑ T₃, ↑ T₄, ↓ TSH. - The thyroid gland itself is over-producing, causing suppression of TSH. - HY example is Graves disease (activating-antibody against TSH receptor), toxic adenoma (autonomously secreting nodule), and toxic multinodular goiter (multiple autonomously secreting nodules). - TSH is suppressed due to ↑ negative-feedback at the hypothalamus / anterior pituitary.
Secondary hyperthyroidism	- ↑ T₃, ↑ T₄, ↑ TSH. - TSH- or TRH-secreting tumor causing over-stimulation of thyroid gland. - Despite the ↑ thyroid hormone, TSH can't be suppressed because there is autonomous secretion from a tumor. - In theory, hypothalamic / anterior pituitary glandular hyperplasia, or inflammation (autoimmune hypophysitis) could also cause ↑ TSH leading to secondary hyperthyroidism.

Exogenous administration physio	
T ₄ administration	- If thyroxine is administered, we will have ↑ T₃, ↑ T₄, ↓ TSH. - T₄ is converted to T₃ peripherally, so if T₄ is ↑ due to exogenous administration, then T₃ is also ↑. - TSH is suppressed due to negative-feedback.
T ₃ administration	- If triiodothyronine is administered, we will have ↑ T₃, ↓ T₄ (oh wow), ↓ TSH. - Remember: T₄ is converted to T₃, but T₃ is not converted to T₄.

	- If we administer T3 exogenously, we will have suppression of TSH due to negative-feedback, leading to ↓ endogenous T4 synthesis. Even though endogenous T3 production is also ↓, since we're administering exogenously, it is clearly still ↑.
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HY Thyroid diagnoses	
Hashimoto thyroiditis	- Autoimmune destruction of thyroid gland. - Lymphocytic infiltrate seen on biopsy (asked on NBME). - Anti-microsomal (aka anti-thyroperoxidase) + anti-thyroglobulin antibodies. - ↓ T3, ↓ T4, ↑ TSH (primary hypothyroidism). - ¹³¹I uptake scan is ↓ or patchy (i.e., the gland is not producing hormone). - Buzzy findings are weight gain, brittle hair, dry/doughy skin, and cold intolerance. The USMLE often will omit these findings because they're too easy. - HY additional findings I've seen show up in NBME Qs are: bradycardia (55-60), menstrual irregularity, dysthymia/depression/apathy, carpal tunnel syndrome (due to glycosaminoglycan [GAG] deposition), ↑ cholesterol, proximal muscle weakness with ↑ creatine kinase (hypothyroid myopathy), transaminitis (sounds weird, but LFTs can be ↑). - Associated with other autoimmune diseases (e.g., type I diabetes, pernicious anemia, vitiligo) and immunodeficiencies (e.g., IgA deficiency). For USMLE, a HY big-picture concept is that "Autoimmune diseases go together," where if you have one autoimmune disease, there's ↑ risk of other autoimmune diseases. The strict HLA associations aren't important (outside of HLA-B27). In addition, "Autoimmune diseases and immunodeficiencies go together," where having autoimmune disease in general ↑ risk of immunodeficiencies, and vice versa. - ↑ risk of non-Hodgkin lymphoma (e.g., primary CNS lymphoma). - Tx = levothyroxine or triiodothyronine.
Graves disease	- Activating autoantibody against TSH receptor. This antibody is called thyroid-stimulating immunoglobulin (TSI). - Lymphocytic infiltrate + hyperplastic thyroid follicles seen on biopsy. - ↑ T3, ↑ T4, ↓ TSH (primary hyperthyroidism). - ¹³¹I uptake scan shows diffusely ↑ uptake. - Exophthalmos (aka proptosis; protrusion of eyes) and pretibial myxedema (non-pitting edema of shins due to GAG deposition) are specific findings in Graves. The vignette need not mention either of these findings, but if the Q specifically tells you "there's no exophthalmos," they're saying it's not Graves. - Buzzy findings are tachycardia, heat intolerance, sweating, and palpitations. - Myopathy can also occur with ↑ CK, but USMLE tends to give this finding in hypothyroidism. - Can cause atrial fibrillation in younger patients (normally AF is elderly). - "Thyroid storm" is an acute exacerbation of Graves, often precipitated by a stress factor such as surgery, trauma, or infection, where thyroid hormone production goes ↑↑↑, causing dangerous physiologic responses, such as hyperthermia, tachycardia, and altered mental status. - Thyroid storm can also cause adrenal crisis (i.e., acute ↓ in blood pressure due to rapid consumption of cortisol), especially in those with concurrent adrenal insufficiency taking exogenous steroids. Tx for the low BP is IV glucocorticoid (i.e., hydrocortisone or methylprednisolone). - Tx for general Graves is thionamides (propylthiouracil [PTU] and methimazole). - PTU and methimazole both inhibit thyroperoxidase; PTU has an additional MOA of inhibiting 5-deiodonise. - Both PTU and methimazole can cause neutropenia/agranulocytosis.

	- Propranolol is used for tachycardia in hyperthyroidism (beta-blockade inhibits 5-deiodonase). - Oral prednisone is specifically used for Tx of exophthalmos, not thionamides. - Tx of thyroid storm on USMLE is tetrad of: 1) PTU, 2) propranolol, 3) steroids, and 4) potassium iodide (shuts off thyroid gland due to Wolff-Chaikoff effect). - Wolff-Chaikoff effect is a transient ↓ in thyroid hormone synthesis in the setting of acute ↑ in iodide exposure, which triggers the thyroid gland to acutely shut off. - Tangentially, if a USMLE Q gives you a patient exposed to radioactive iodine (e.g., working in a laboratory), the Tx is potassium iodide, which saturates the thyroid gland with “regular” iodide and prevents uptake of the radioiodine, thereby preventing destruction of the gland.
deQuervain thyroiditis	- Aka subacute granulomatous thyroiditis, or just simply subacute thyroiditis. - Mechanism is viral infection followed by a painful/tender thyroid. There is inflammation of the thyroid gland, which causes the spacing between the cells to increase slightly, allowing for the release of pre-formed thyroid hormone into the blood. Therefore we have ↓ TSH, ↑ T3, ↑ T4, ↓ ¹³¹I uptake. - The gland is not over-producing thyroid hormone. This is why uptake is not increased. - Subacute granulomatous thyroiditis can be either hypo- or hyperthyroid. The key detail you need to know is that uptake is always decreased even if the patient is hyperthyroid. - DeQuervain vignettes will almost always be given as hyperthyroidism because the USMLE wants to specifically assess that you know uptake is decreased. If they give you hypo-, of course you’ll select decreased for uptake. - Decreased uptake applies to all thyroiditis conditions (i.e, deQuervain, drug-induced, and postpartum). - Viral infections are often asymptomatic, so most deQuervain vignettes will not mention the viral infection. They’ll just mention a tender/painful thyroid.
Drug-induced thyroiditis	- Caused by lithium or amiodarone on USMLE. - Can in theory be hypo-, eu-, or hyperthyroid. - Will usually present on USMLE as painless hypothyroidism in patient being treated for bipolar disorder (lithium) or started on new anti-arrhythmic (amiodarone). - Will present with ↓ ¹³¹I uptake, where inflammation of the gland can sometimes cause leakage of thyroid hormone into the blood, but the gland itself is not demonstrating increased production of hormone.
Subclinical hypothyroidism	- Elevation in TSH despite normal thyroid hormone levels. That is: - ↑ TSH, ↔ T3, ↔ T4. - Is thought to be a premonition to Hashimoto in some patients. - In subclinical hypothyroidism, the patient will be asymptomatic (hence subclinical) and will have normal T3 and T4, despite an elevated TSH. - Most patients with subclinical do not need to be treated. - 2CK/3: Don’t treat unless TSH >10, patient is pregger, or Hashimoto antibodies are (+).
Euthyroid sick syndrome	- Vignette will give patient being weaned from a ventilator, or someone who’s had recent major trauma or surgery. - ↔ TSH, ↔ T4, ↓ T3, ↑ rT3. - Etiologies are manifold, but one proposed mechanism is that in times of stress (usually acute), spikes in cortisol and other inflammatory mediators can inhibit peripheral conversion of T4 to T3. This causes T3 to go down. - T4, however, rather than being detectable in excess, is converted to an inactive form of thyroid hormone called reverse T3 (rT3), so rT3 is high. - T4 is therefore still in the normal range, as is TSH (presumably because T4 is normal).

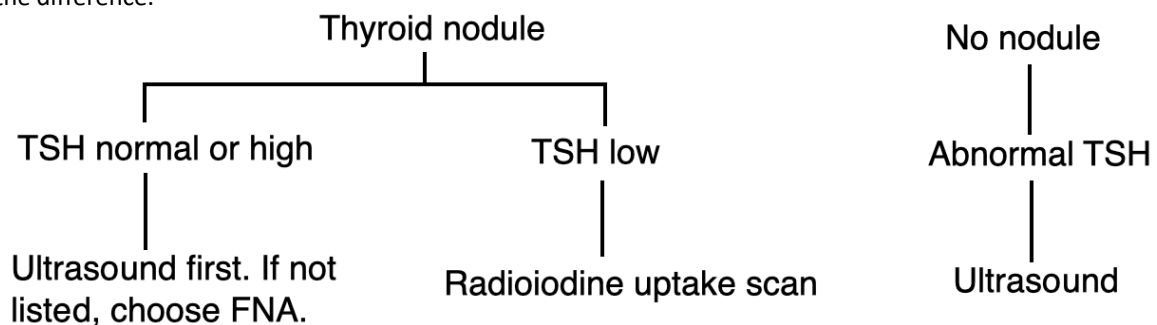
	- Call it weird all you want, but USMLE asks this. This is one of the most “underrated” thyroid diagnoses, as students will often have heard of it but then disregard it, thinking it’s minutiae and probably won’t show up. - One NBME Q gives a patient who had recent major surgery + now has normal TSH and low T3 + doesn’t even mention T4 or rT3.
Toxic adenoma	- Autonomously secreting thyroid nodule (aka hot nodule). - ¹³¹I scan shows uptake onto into the area of a single nodule. - ↑ T3, ↑ T4, ↓ TSH. - Not malignant (i.e., no metastatic potential). - Hot nodules on USMLE are benign. If the nodule is cold (i.e., not secreting thyroid hormone), that’s when we worry about cancer (i.e., papillary, follicular, medullary, etc.).
Toxic multinodular goiter	- Autonomously secreting thyroid nodules (i.e., many hot nodules). - ¹³¹I scan shows multinodular uptake. - ↑ T3, ↑ T4, ↓ TSH. - Not malignant (i.e., no metastatic potential). - I don’t think I’ve ever seen this as a correct answer on USMLE, but it shows up quite a bit as a distractor, e.g., for deQuervain Qs. - Occurs in elderly. You might get a younger patient with hyperthyroidism due to deQuervain, and you can eliminate toxic multinodular goiter because 1) the patient is young, and 2) deQuervain is painful/tender, whereas TMG is not.
Riedel thyroiditis	- Obscure fibrosing disorder characterized by replacement of the thyroid gland with dense, fibrous tissue (sometimes called “woody thyroid” for this reason). - Can sometimes resemble anaplastic thyroid carcinoma, where it extends into adjacent structures such as the esophagus or trachea.
Factitious thyrotoxicosis	- Aka surreptitious thyrotoxicosis. - The patient is injecting thyroid hormone. - USMLE need not mention the patient is a pharmacist (implying access to the hormone). - Q will give hyperthyroid patient with small, non-palpable thyroid gland (atrophic due to suppressed TSH). - As mentioned earlier, T4 is converted to T3, but T3 isn’t converted to T4. - So if thyroxine is administered, we will have ↑ T3, ↑ T4, ↓ TSH. - If triiodothyronine is administered, we will have ↑ T3, ↓ T4, ↓ TSH.
Struma ovarii	- Abstruse ovarian tumor that for some magical reason secretes T3 and T4.

Thyroid cancers	
- All malignant (i.e., have metastatic potential). - Can all present as solitary cold nodules in the neck, especially if Hx of radiation exposure (e.g., Hodgkin). - Thyroid cancer will always be a cold nodule (i.e., non-secretory) on USMLE – i.e., the patient will be either euthyroid or hypothyroid. If a nodule is hot (i.e., demonstrates uptake), it is instead a toxic adenoma, which is not malignant.	
Papillary	- The answer if they say “branching papillae with fibrovascular cores” – i.e., looks like villi on light microscopy. - Has psammoma bodies (rings of calcium). - Orphan Annie nuclei are a garbage histo finding. Highly overrated – i.e., students somehow remember this detail above all others for USMLE even though its yieldness is nonexistent. - Spreads lymphatically.
Follicular	- Basically nonexistent thyroid cancer on USMLE. - Known as the “least exciting” thyroid cancer because it just looks like thyroid follicles. - Spreads hematogenously.
Medullary	- Highest yield thyroid cancer on USMLE. - Has amyloid on light microscopy, which demonstrates apple-green birefringence with Congo red stain.

	- USMLE wants you to know the amyloid is β-pleated sheets (i.e., as opposed to α-helices). - Associated with <i>RET</i> proto-oncogene mutations. - Serum calcitonin is $\uparrow$ due to either direct secretion by the cancer or parafollicular C cell hyperplasia (i.e., the C cells of the thyroid gland make calcitonin). You should also know the MOA of calcitonin is inhibition of osteoclast activity (i.e., it does not put calcium back into the bone from the blood; it merely caps off the calcium that's already in the blood by preventing more from leaching out of the bone). - <i>RET</i> mutations can cause isolated familial medullary thyroid cancer, or it can cause full-on MEN-2A/B syndromes (discussed later). The point is, <i>RET</i> mutations need not cause the other findings in the MEN-2 syndromes; the Q can just give isolated medullary thyroid cancer in a family, and the answer is <i>RET</i> mutation.
Anaplastic	- Rare; occurs in elderly; invades adjacent structures such as the esophagus and trachea.

Thyroid nodule evaluation (2K/3 only)

- When evaluating for thyroid cancer, the first step is "palpation of thyroid gland" before "check serum TSH." There is an NBME Q for 2CK where both are listed and palpation is correct (i.e., do a history and exam before jumping into tests).
- Once a nodule is palpated, *then* serum TSH is the next best step.
- If patient with nodule has normal or high TSH (i.e., is euthyroid or hypothyroid), do ultrasound before fine-needle aspiration (FNA). This is assessed on the new 2CK Free 120, where ultrasound is correct over FNA for evaluation of thyroid nodule. It had long been pushed in resources that ultrasound is always wrong. But this is on Free 120.
- 2CK IM CMS Q gives euthyroid patient with nodule where FNA is the answer, but ultrasound isn't listed.
- So my conclusion (based on NBME/Free 120 content; not my opinion) is: for eu- or hypothyroid patient with thyroid nodule, ultrasound is correct over FNA if both are listed. If ultrasound isn't listed, choose FNA.
- If patient has low TSH (i.e., is hyperthyroid), do ^{131}I uptake scan, not ultrasound.
- Since carcinomas are non-secretory of thyroid hormone, if a patient is hyperthyroid, we're not concerned about carcinoma, which is why we don't go the ultrasound then FNA route. We just do uptake to better see if the patient's etiology for hyperthyroidism is Graves (diffuse), toxic adenoma (single nodular uptake), or toxic multinodular goiter (multifocal nodular uptake).
- A caveat about the above algorithm for nodule evaluation is that if a patient has hyperthyroidism (i.e., low TSH) *and does not have a nodule*, then you will choose ultrasound over uptake scan. I know this sounds annoying, but USMLE assesses this. There is a new NBME Q where they give low TSH in young patient with no mention of nodule, and the answer is ultrasound over uptake scan. But in Qs where TSH is low and the patient has a nodule, the answer is uptake scan first over ultrasound. If you want a high score, then know the difference.



- TSH is screening for everyone not pregnant.
- T3+T4 can be ordered after TSH, but Q often omits this.
- Screening in pregnancy is Free-T4.
- Most diagnostic/accurate for everyone is Free-T4.
- Algorithms that are more detailed as far as size of nodule, etc., are basically nonsense for USMLE.

Thyroid in pregnancy

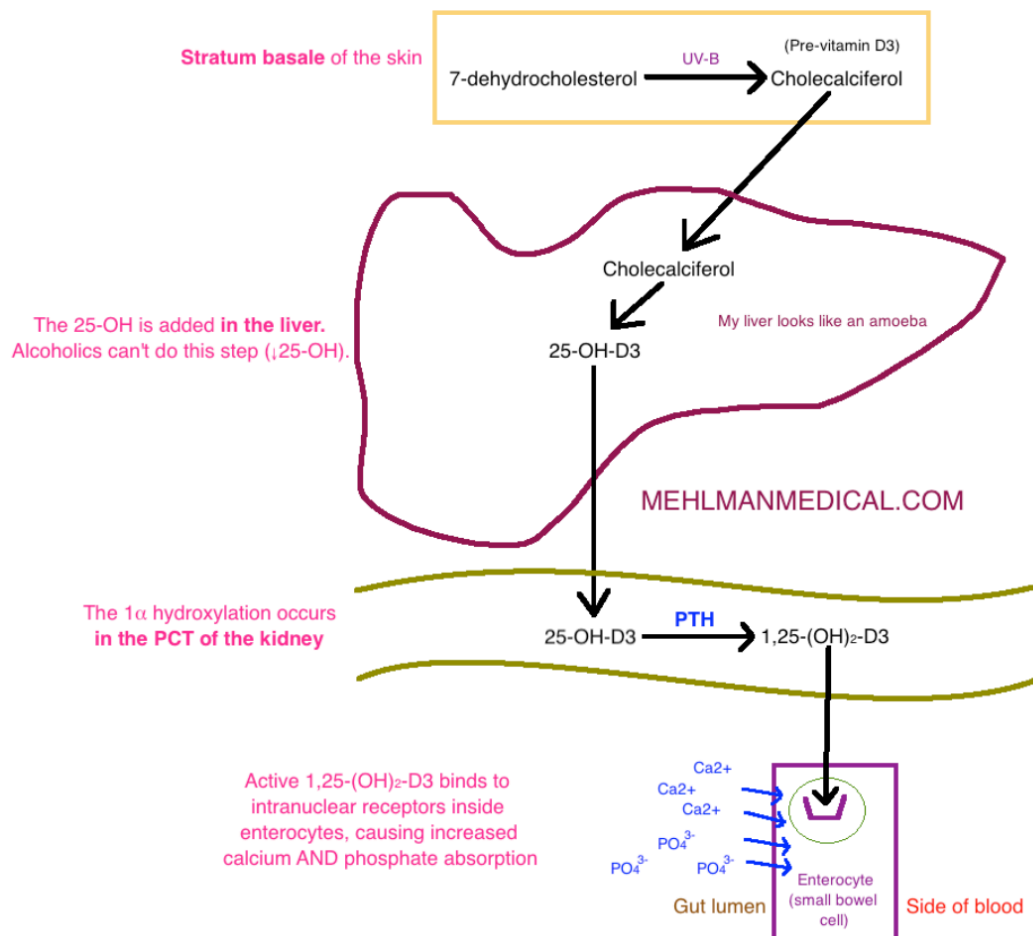
- For pregnancy on USMLE, choose the combo of **no change TSH, no change free-T4, ↑ total T4** for women who have no thyroid symptoms.
- **Estrogen** causes ↑ thyroid-binding globulin (TBG) production by the liver. TBG is the protein carrier molecule for thyroid hormone in the blood. An NBME Q asks for which hormone causes the ↑ TBG in pregnancy → answer = estrogen; progesterone is wrong answer.
- Free-T4 is the physiologically active form of thyroid hormone. T4 protein-bound to TBG (99%) has minimal effect. Free-T4 + TBG-bound T4 = total T4.
- TBG will mop up free-T4, causing free-T4 to transiently decrease and TSH to rise (less negative feedback). This rise in TSH will stimulate more production of T4 by the thyroid gland, making total T4 go up. The absolute amount of free-T4 will increase back to normal, thereby suppressing TSH back to normal. But the total amount of T4 is now increased – i.e., free-T4 is normal again, but TBG-bound T4 is higher.
- T3 is normal because free-T4 is normal. Free-T4 is peripherally converted to T3. I've never seen anything about "free-T3" on NBME material and I wouldn't worry about it.
- A student might say, "Wait, but why are you giving the above bold arrows if you just gave me all sorts of transient changes in the arrows based on TBG?." It's because the bold arrows are what the USMLE wants. Pregnant women who are euthyroid will have normal free-T4 and increased total T4, and their TSH will be normal. The changes due to TBG rising are likely synchronous and slow enough that the patient's TSH and free-T4 stay within reference ranges.
- Postpartum (silent) thyroiditis can result in either hypo- or hyperthyroidism following parturition. These arrows are unrelated to the aforementioned ones. The highest yield point you need to know is that ¹³¹I uptake into the thyroid gland is low, even if the patient is hyperthyroid. This is the same for deQuervain and drug-induced thyroiditis, where uptake is always low. This is because with thyroiditis conditions, there is merely increased spacing between the cells of the thyroid gland due to inflammation, allowing thyroid hormone to leak out into the blood. The gland itself is not excessively producing thyroid hormone. Then we have negative feedback causing low TSH, and in turn less stimulation of the thyroid gland, which is why uptake is low.
- If USMLE asks you about levothyroxine dosing during pregnancy, the answer is "increase dose by 50%."
- Avoid methimazole in first trimester (teratogenic).
- PTU is the correct answer for thyroid storm during pregnancy, even though longer-term use in 2nd and 3rd trimesters isn't considered ideal because of hepatic toxicity risk. In other words, NBME will give 3rd trimester woman with thyroid storm, where PTU is part of the treatment, and student is confused because they thought we avoid that in 2nd and 3rd trimesters, but it's what NBME wants.

Thyroid in pediatrics

Thyroglossal duct cyst	- Benign, painless, midline neck mass in a school-age kid that arises from a persistent thyroglossal duct (i.e., the embryologic remnant of the thyroid gland's descent from the base of the tongue to its final position in the neck). This point of origin at the base of the tongue is called the foramen cecum. - 2/3 of Qs will give a painless midline neck lump that moves upwards with swallowing or protrusion of the tongue (due to its attachment to the hyoid bone and/or the base of the tongue). - 1/3 of Qs won't mention the buzzy upward movement with swallowing or protrusion of the tongue; instead, it will just say a kid has a painless midline neck mass just inferior to the hyoid bone that demonstrates ⁹⁹Tc uptake. - On USMLE, the answer = "endoderm of foramen cecum," when they ask about embryo.
Lingual thyroid	- Obscure cause of hypothyroidism that shows up on an offline NBME. - Refers to presence of ectopic thyroid tissue located at the base of the tongue due to failure of the thyroid gland to descend from the foramen cecum. - This is different from thyroglossal duct cyst, where this ectopic tissue may be the only functioning thyroid tissue in the body, resulting in hypothyroidism.

	- The USMLE Q will say a kid has hypothyroidism + a midline neck lump located high in the neck. They say nothing about protrusion of the tongue or uptake into the mass. - Can sometimes cause dysphagia (trouble swallowing), dysphonia (voice changes), or dyspnea (difficulty breathing).
Cretinism	- Aka congenital hypothyroidism. - Most common causes are iodine deficiency in the mother during pregnancy (worldwide) and fetal thyroid dysgenesis (western countries). - Leads to mental retardation (poor myelin sheath development), impaired bone growth, hypotonia, macroglossia, and protuberant abdomen (due to umbilical hernia; can be confused with kwashiorkor, which is protein-calorie malnutrition causing ascites). - Screened for at birth using the heel-prick test to prevent mental retardation.
TBG deficiency	- The USMLE does not expect you to know about some obscure condition called thyroid-binding globulin deficiency. The reason they ask about this is because the role of TBG on thyroid hormones in pregnancy is exceedingly high-yield, so if you know the reasoning/mechanism behind the latter, you can easily infer what would be seen in TBG deficiency (i.e., the inverse). - $\leftrightarrow$ TSH, $\leftrightarrow$ T3, $\leftrightarrow$ free-T4, $\downarrow$ total T4. - In pregnancy, since TBG is high, this ultimately results in high total T4 despite a normal free-T4. So the student can easily infer, "Well, if our TBG merely is low, rather than high, then total T4 must be low while free is same." Sort of like pregnancy but just the opposite direction.

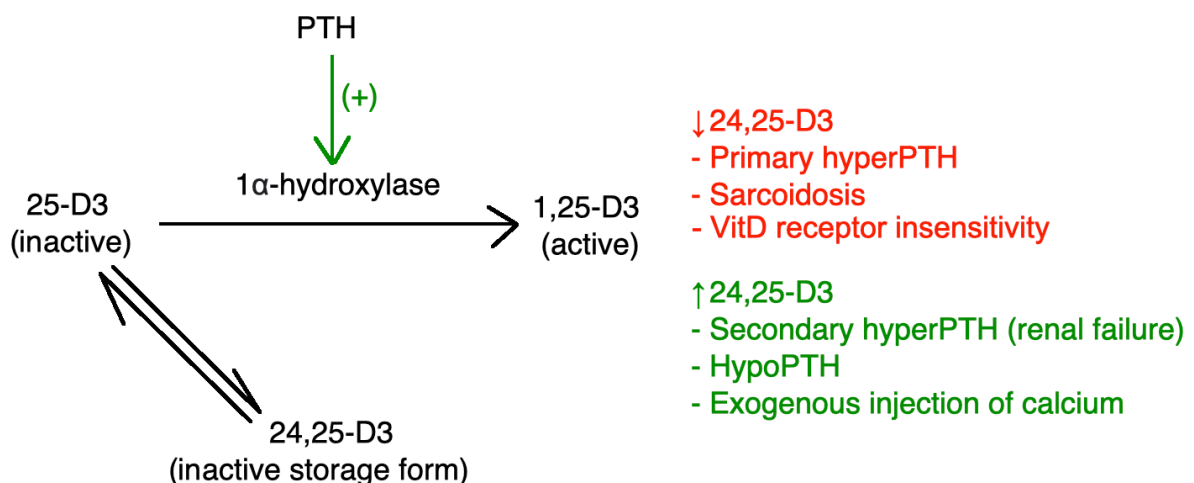
Vitamin D synthesis for the USMLE



Parathyroid intro points	
Blood supply	- Same as blood supply to the thyroid glands (i.e., the inferior and superior thyroid arteries). - This is relevant because post-thyroidectomy, disrupted blood supply and/or removal of the parathyroid glands results in ↓ serum Ca^{2+} (discussed later).
Embryo/Anatomy	- Four parathyroid glands located on the back of the thyroid gland. Not complicated.
Physio/function	- Parathyroid hormone (PTH) functions to ↑ serum Ca^{2+} and ↓ serum PO_4^{3-}. - Parathyroid hormone (PTH) functions to increase serum calcium in three main ways: 1) Pulls calcium out of the bone and puts it in the blood. ○ PTH binds to osteoblasts, which will then express RANK-L on their cell surface, which will bind to RANK receptor on the surface of osteoclasts. Osteoclasts then resorb (break down) bone. The calcium enters the blood, increasing serum calcium. 2) Increases renal reabsorption of calcium. ○ PTH causes Ca^{2+} reabsorption at the late-DCT of the kidney by increasing expression of an apical calcium channel. 3) Upregulates 1α-hydroxylase. ○ PTH upregulates 1α-hydroxylase in the PCT of the kidney, which converts inactive 25-OH-vitamin D3 (calcidiol) into active $1,25\text{-(OH)}_2\text{-D}_3$ (calcitriol). ○ $1,25\text{-(OH)}_2\text{-D}_3$ then goes to the small bowel, where it increases absorption of both calcium and phosphate. - PTH decreases serum phosphate by inhibiting its reabsorption in the PCT of the kidney. ○ PTH causes the downregulation of apical PCT phosphate transporters. ○ If PTH is low, we don't get downregulation of these transporters, too much phosphate is reabsorbed, and serum PO_4^{3-} goes up.

Vitamin D pathologies	
Rickets/Osteomalacia	- ↓ Serum Ca^{2+}, ↓ serum PO_4^{3-}, ↑ PTH. - Rickets = vitamin D deficiency in children. - Osteomalacia = vitamin D deficiency in adults. - In classic rickets/osteomalacia caused by dietary deficiency, lack of sunlight, or liver disease, calcium and phosphate are both low because vitamin D is needed for intestinal absorption of both. - PTH goes up because calcium is low (decreased negative feedback at the calcium-sensing receptors of the parathyroid glands). - It should be noted that in vitamin D deficiency caused by renal failure, phosphate is high, not low, because the effect of the renal failure on phosphate levels "wins" over the mere vitamin D deficiency. In renal failure, the kidney cannot downregulate the PCT phosphate reabsorption channels, thereby increasing absorption. - Never choose low phosphate in renal failure, even though vitamin D3 activation is low. - Renal failure causing low vitamin D: ○ ↓ serum Ca^{2+}, ↑ serum PO_4^{3-}, ↑ PTH. - No renal failure + low vitamin D (i.e., classic rickets/osteomalacia): ○ ↓ serum Ca^{2+}, ↓ serum PO_4^{3-}, ↑ PTH.
Inadequate sunlight	- UV-B radiation is necessary to convert 7-dehydrocholesterol into cholecalciferol in the basal layer of the skin. - The USMLE wants you to know that 7-dehydrocholesterol is made even in those who don't get adequate sunlight. It is <i>cholecalciferol</i> that isn't produced if patient doesn't receive any sun exposure.

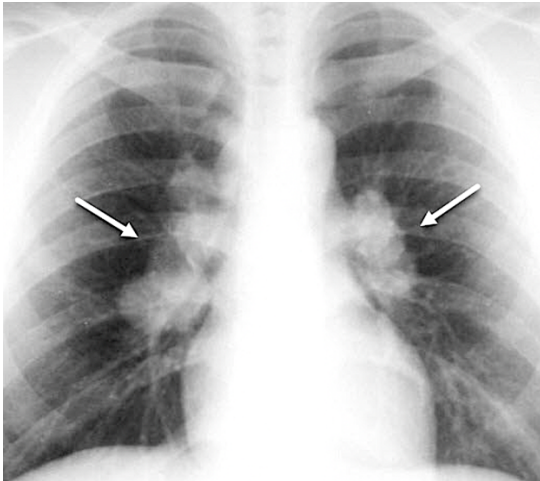
	- There is NBME Q where they give vitamin D deficiency in patient with lack of sunlight, and they ask for which substrate's synthesis is directly decreased, and the answer is cholecalciferol; 7-dehydrocholesterol is wrong answer.
Alcoholism	- Alcoholism implies hepatic insufficiency. - The patient will have decreased synthesis of 25-D3. - That means the patient will have ↓ 25-D3 as well as ↓ 1,25-D3.
Renal failure	- Inability for PTH to activate 1α-hydroxylase in the PCT of the kidney. - ↓ 1,25-D3 synthesis, but 25-D3 is normal. - 25-D3 doesn't build up physiologically. If there is "extra" 25-D3 because it is not being converted to 1,25-D3, then it gets shunted sideways to another inactive form called 24,25-D3 (asked on NBME). - Osteomalacia resulting from renal failure is called renal osteodystrophy. - There is an NBME Q where they give a non-displaced fracture (pseudofracture) of the femoral head in a patient with renal failure + they ask the mechanism; answer is "decreased bone mineralization," since Vit D is necessary to convert unmineralized osteoid into mineralized hydroxyapatite.
Vitamin D receptor insensitivity	- ↓ Serum Ca^{2+} (↓ bowel absorption if Vit D can't do its job). - ↑ PTH (goes ↑ to compensate since Ca^{2+} is low). - ↓ 25-D3 (more converted to 1,25-D3 since PTH is ↑). - ↓ 24,25-D3 (since less 25-D3 available). - The NBME gives normal serum levels for $1,25-(\text{OH})_2\text{-D}_3$. This might sound weird, since more 1,25-D3 should be produced to compensate. But perhaps in patients with this condition, 1,25-D3 is often normal, rather than overtly elevated. - The NBME doesn't ask about phosphate levels for this condition, possibly because whilst vitamin D deficiency will ordinarily cause low phosphate (due to decreased intestinal absorption), it is possible that in patients with chronically elevated PTH in the setting of a vitamin D receptor mutation, phosphate might be variable. - An NBME Q mentions 24,25-D3 levels as "undetectable" in the setting of vitamin D receptor mutation. This also means if USMLE asks you for merely the 25-OH-D3 levels, choose a down arrow.



Hyperparathyroidism conditions	
Primary	- $\uparrow \text{Ca}^{2+}$, $\downarrow \text{PO}_4^{3-}$, $\uparrow \text{PTH}$. - Primary hyperparathyroidism is usually due to a PTH-secreting adenoma, but can also be due to diffuse four-gland hyperplasia. - Can be part of MEN-1 or -2A (discussed later). - $\uparrow$ Urinary Ca^{2+}, $\uparrow$ serum Ca^{2+}, $\uparrow$ urinary cAMP. - Some students will immediately think there's an erratum with the above arrows. There's not. Relax. - In primary hyperparathyroidism, PTH will be high. PTH functions to reabsorb calcium in the late-DCT of the kidney, so in theory, you'd think urinary calcium would be low, not high. Sounds logical. - However, even though more Ca^{2+} is indeed reabsorbed in the late-DCT, since serum Ca^{2+} is high in primary hyperparathyroidism, <i>more calcium is filtered through the glomerulus to begin with</i>. So even though more is reabsorbed distally, the urinary calcium <i>is still high</i> because we had more to start with. - PTH causes increased urinary cAMP. This is a weird variable that you should be aware of. PTH acts through G-α-s G-proteins, where adenylyl cyclase activity and, in turn, cAMP are increased.
Secondary	- $\downarrow \text{Ca}^{2+}$, $\uparrow \text{PO}_4^{3-}$, $\uparrow \text{PTH}$, $\downarrow 1,25\text{-D}_3$. - Secondary hyperparathyroidism is due to renal failure on USMLE. - Technically the term can be applied to ordinary rickets/osteomalacia, when PTH goes up to compensate for low serum calcium. But as I said, you should think of secondary hyperparathyroidism as basically just synonymous with renal failure causing low calcium, where PTH goes up to compensate. - There are two reasons why calcium is low: 1) The kidney cannot reabsorb it in the late-DCT like it's supposed to. 2) The kidney cannot synthesize activated 1,25-D₃ in the PCT like it's supposed to. Since 1,25-D₃ absorbs calcium in the small bowel, and 1,25-D₃ levels are lower, serum calcium will also be lower. - The low serum calcium will cause PTH to go up (deceased negative-feedback at the calcium-sensing receptors on the parathyroid glands). - Since the etiology for the increased PTH secretion is not the parathyroid glands themselves, we call this secondary hyperparathyroidism. If the etiology is due to the parathyroid glands themselves (i.e., adenoma or diffuse hyperplasia), we call that primary hyperparathyroidism. - The calcium and phosphate arrows for secondary hyperPTH are the opposite of primary hyperPTH. - Phosphate is high in renal failure, despite the high PTH, because the kidney is not able to downregulate the PCT phosphate transporters, so there is too much phosphate reabsorption.
Tertiary	- $\uparrow \text{Ca}^{2+}$, $\uparrow/\leftrightarrow \text{PO}_4^{3-}$, $\uparrow \text{PTH}$. - In order to understand tertiary hyperparathyroidism, let's first review secondary hyperparathyroidism: - Patients with renal failure will initially develop secondary hyperparathyroidism, where calcium is low, phosphate is high, and PTH is high. Calcium is low due to failure of reabsorption at the late-DCT of the kidney and because of decreased synthesis of activated vitamin D₃ (leading to decreased small bowel absorption). Phosphate is high due to failure to downregulate PCT reabsorption pumps (i.e., more pumps reabsorb phosphate). PTH is high because calcium is low (less negative feedback). - Tertiary hyperparathyroidism results from hyperplasia of the parathyroid glands in patients with long-standing secondary hyperparathyroidism, such that even if the renal failure is brought under control and serum calcium is brought back into the normal range, PTH continues being autonomously secreted at higher basal levels than prior to the renal disease, effectively resetting the body's setpoint for calcium homeostasis.

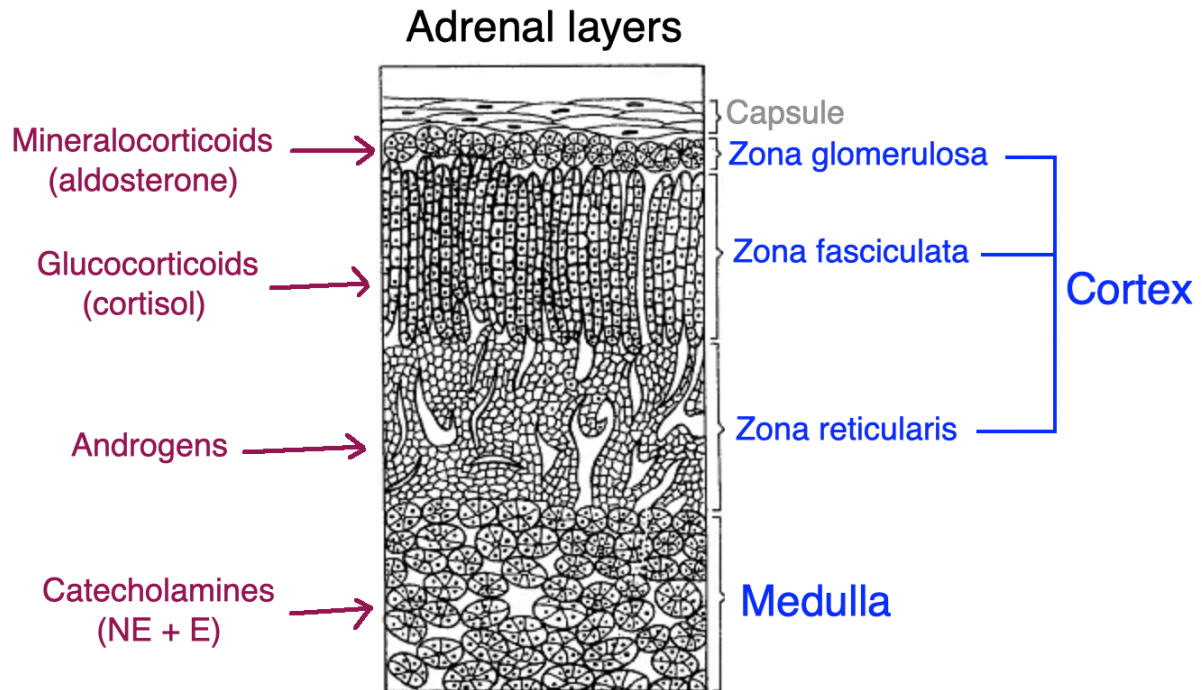
	- This causes a rise in serum calcium in a patient with renal failure. This should immediately raise a red flag for tertiary hyperPTH, since renal failure patients will almost always have low, not high, calcium. - Even though the parathyroid glands will be hyper-secreting PTH, this is not primary hyperparathyroidism, since the etiology for the high PTH was not idiopathic adenoma or hyperplasia. - Phosphate can be high because the patient still has renal insufficiency, even if controlled.
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Hypoparathyroidism conditions	
DiGeorge	- $\downarrow \text{Ca}^{2+}$, $\uparrow \text{PO}_4^{3-}$, $\downarrow \text{PTH}$, $\downarrow \text{T cell levels}$, $\leftrightarrow \text{B cell levels}$. - Mechanism is most frequently 22q11 deletion, resulting in agenesis of the 3rd and 4th pharyngeal pouches. - The 3rd pouch becomes the thymus + two inferior parathyroids. - The 4th pouch becomes the two superior parathyroids. - Tetralogy of Fallot and truncus arteriosus are common heart defects. - Agenesis of the parathyroid glands $\rightarrow$ primary hypoparathyroidism $\rightarrow$ low calcium + high phosphate. - DiGeorge is characterized by T cell deficiency. Recurrent viral, fungal, and protozoal infections are characteristic. - Absent thymic shadow = T cell deficiency (DiGeorge and SCID). - Scanty lymph nodes/tonsils = B cell deficiency (Bruton XLA and SCID).
Post-thyroidectomy	- $\downarrow \text{Ca}^{2+}$, $\uparrow \text{PO}_4^{3-}$, $\downarrow \text{PTH}$. - Thyroidectomy can result in removal of or damage to the parathyroid glands, resulting in decreased PTH secretion and, in turn, hypocalcemia. Phosphate goes up because there is increased renal reabsorption (PTH normally downregulates apical PCT phosphate transporters, thereby promoting excretion). - Hypocalcemia causes "up" findings – i.e., muscle tetany and hyperreflexia. - Twitching of the masseter with stimulation is called Chvostek sign of hypocalcemia. - Trousseau sign of hypocalcemia is carpopedal spasm with blood pressure cuff inflation (i.e., twitching of the hand/wrist). The USMLE can give you either Chvostek or Trousseau sign when Ca^{2+} is low. - Hypercalcemia, in contrast, causes "down" findings – i.e., muscle flaccidity and hyporeflexia. - Do not confuse Trousseau sign of hypocalcemia with Trousseau and Troisier signs of malignancy. - Troisier sign of malignancy is a palpable left supraclavicular lymph node sometimes seen in visceral malignancy (aka Virchow node). - Trousseau sign of malignancy is migratory thrombophlebitis classically seen in head of pancreas adenocarcinoma (but can be other adenocarcinomas, such as bronchogenic).
Hypomagnesemia	- $\downarrow \text{Ca}^{2+}$, $\uparrow \text{PO}_4^{3-}$, $\downarrow \text{PTH}$, $\downarrow \text{Mg}^{2+}$. - Alcoholics are susceptible to hypomagnesemia as a result of dietary deficiency (EtOH is 7kcal/g, so they fill up on alcohol). - Low magnesium can cause hypocalcemia and hypokalemia nonresponsive to supplementation (i.e., you give calcium or potassium for low serum levels, but the serum levels do not appreciably rise). - Basal levels of magnesium are required for proper functioning of the parathyroid gland, so low magnesium can cause low PTH secretion. - Patients with hypomagnesemia may indeed have completely normal serum levels for calcium, phosphate, and PTH, but if we are forced to choose arrows, we would choose a $\text{Ca}^{2+}/\text{PO}_4^{3-}/\text{PTH}$ combo that reflects low PTH secretion. - The USMLE will just say an alcoholic has low calcium or potassium nonresponsive to supplementation, and then the next best step in management is "check serum magnesium levels." This isn't hard at all. But some students will be like "Oh wow."

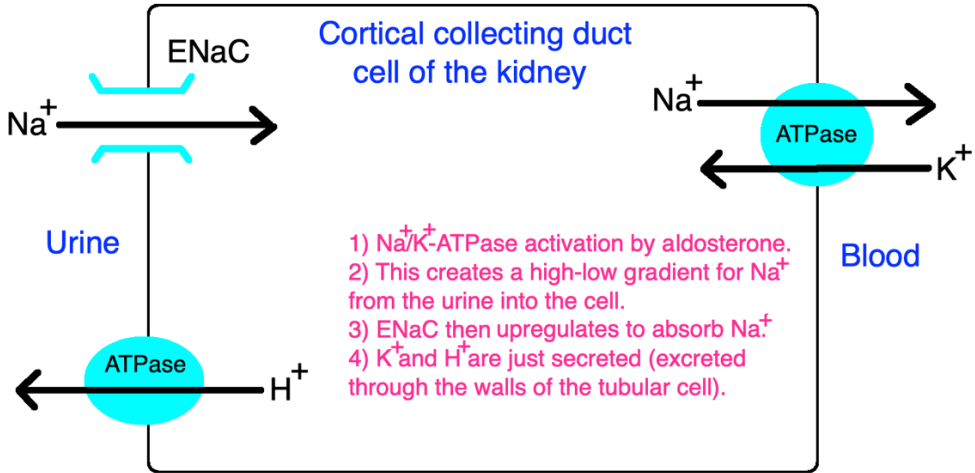
Other calcium derangement conditions	
Sarcoidosis	- Idiopathic autoimmune disorder that is one of the highest yield conditions on USMLE. You must know this condition extremely well. - Characterized by non-caseating granulomas within lung tissue. These consist of activated macrophages called epithelioid macrophages, aka histiocytes. - These histiocytes in the lung secrete 1α-hydroxylase (normally produced in the PCT of the kidney in response to PTH), which will convert inactive 25-OH-D3 into active 1,25-(OH)$_2$-D3 (i.e., causing high vitamin D; aka hypervitaminosis D), which then goes to the small bowel and $\uparrow$ absorption of calcium, causing hypercalcemia. - Even though 1,25-(OH)$_2$-D3 also $\uparrow$ small bowel absorption of phosphate along with calcium, it's to my observation that phosphate can be <i>normal</i> in sarcoidosis NBME Qs, so do not be confused if you see phosphate in the normal range and only see hypercalcemia. - PTH is <i>suppressed</i> in sarcoidosis due to the high Ca^{2+} (exceedingly HY). - In other words, on USMLE you will select $\uparrow$ 1,25-(OH)$_2$-D3 and $\downarrow$ PTH. - Archetypal presentation is African-American woman 20s-30s with 6+ months of dry cough and red shins (erythema nodosum). Other findings like low-grade fever with flares can occasionally be seen. - Bihilar lymphadenopathy seen on CXR or CT. They can also describe this as CXR or CT "shows hilar nodularity."  - There is a Family Med Q that gives early-30s African-American woman with dry cough + they say CXR shows no abnormalities $\rightarrow$ answer = "activation of mast cells" (i.e., asthma is the answer); wrong answer is "non-caseating granulomatous inflammation." So your HY point here is: if the vignette sounds like it could be sarcoidosis but they say CXR is normal or shows "mild hyperinflation" (also buzzy for asthma), choose asthma over sarcoidosis. - A 2CK IM CMS form gives a sarcoidosis vignette where they mention bihilar lymphadenopathy on a CT scan + high hepatic AST + ALP + weight loss; they don't mention hypercalcemia. Unusual combo of findings, but the answer is inferable based on the CT. Sometimes there can be bile ductule involvement in sarcoidosis. - Lupus pernio is an enlarged nose due to sarcoidosis (not SLE, despite the name). - USMLE wants steroids (i.e., oral prednisone) as treatment.
Paraneoplastic PTHrp	- Squamous cell carcinoma of the lung and renal cell carcinoma both secrete parathyroid hormone-related peptide (PTHrp). <i>This is not the same as endogenous PTH.</i> - USMLE Qs for renal cell carcinoma and squamous cell of the lung will have:

	○ $\uparrow$ Serum Ca^{2+}, $\downarrow$ serum PO_4^{3-}, $\downarrow$ PTH. - PTHrp exerts very similar physiologic effects as PTH, causing high serum calcium and low phosphate. - PTH secretion by the parathyroid glands will be suppressed in this setting due to the high serum calcium. - NBME will give you lung cancer + high serum calcium + low serum PTH, with the answer being squamous cell carcinoma of the lung. The low PTH will throw some people off. But this is not weird. PTH is not the same as PTHrp (which will be up-arrow).
Metastases	- $\uparrow$ Serum Ca^{2+}, $\downarrow$ PTH. - Once metastases seed at bony locations, cytokine activity causes lysis of the bone and release of calcium into the blood. Hypercalcemia is common in the setting of metastatic malignancies. - PTH is suppressed due to the high calcium.
Paget disease	- $\leftrightarrow$ Serum Ca^{2+}, $\leftrightarrow$ serum PO_4^{3-}, $\leftrightarrow$ PTH, $\uparrow$ ALP. - Idiopathic disorder of increased bone turnover. Bone is described as having mixed osteoblastic and -clastic phases, appearing heterogenous on x-ray. - Buzzy vignette is male over 50 who's hat doesn't fit him anymore + has tinnitus (narrowing of acoustic foramina). - 19/20 questions will give isolated increase in serum ALP. You need to know Ca^{2+}, PO_4^{3-}, and PTH are all normal in Paget disease. There is one Q on a 2CK CMS form where ALP is given as not elevated, but it's a one-off Q and rare. - High-output cardiac failure can occur due to intraosseous AV-fistulae, where patient has an S3 heart sound with high, rather than low, ejection fraction.
Familial hypocalciuric hypercalcemia	- $\uparrow$ Serum Ca^{2+}, $\downarrow$ urinary Ca^{2+}, $\downarrow$ parathyroid Ca^{2+}-sensing receptor sensitivity, $\downarrow$ renal Ca^{2+}-sensing receptor sensitivity. - As the name of the condition implies, for familial hypocalciuric hypercalcemia (FHH), serum calcium will be high and urinary calcium low. - The condition is caused by slightly decreased (not absent) sensitivity of the calcium-sensing receptors at both the parathyroid glands and kidneys, resulting in a slightly higher setpoint for calcium levels in the blood, that may be variable depending on the patient. - Most patients are asymptomatic, and PTH is often higher end of normal, or slightly elevated. Patients do not experience signs of bone resorption sometimes seen in patients with primary hyperparathyroidism. - In primary hyperparathyroidism, despite increased distal renal reabsorption of calcium, urinary calcium is still high because serum calcium is high, so more calcium will be filtered through the glomerulus. In other words, if the urine starts off with more calcium, the amount urinated out is still higher despite increased reabsorption. The calcium-sensing receptors in the renal PCT and loop of Henle function normally and can sense the high calcium, thereby decreasing paracellular reabsorption (unrelated to PTH's effect in the DCT); this allows for greater calciuresis (urination of calcium). - In familial hypocalciuric hypercalcemia, calcium-sensing receptors in the renal tubules (unrelated to PTH) have decreased sensitivity. Therefore, the tubules interpret urinary calcium to be low and will increase reabsorption in the PCT and loop of Henle to compensate, thereby decreasing calciuresis (less urinary calcium). In other words, in FHH, the kidney thinks, "Fuck, we must have low serum Ca^{2+} since I can't sense it, so I'm going reabsorb more so we can retain it."
PseudohypoPTH	- Pseudohypoparathyroidism is caused by a defective PTH receptor, and therefore insensitivity to PTH. - We see $\downarrow$ Serum Ca^{2+}, $\uparrow$ serum PO_4^{3-}, $\downarrow$ urinary cAMP. - If the USMLE asks what change will occur with exogenous PTH administration, choose: $\leftrightarrow$ Serum Ca^{2+}, $\leftrightarrow$ serum PO_4^{3-}, $\leftrightarrow$ urinary cAMP. - There are different types of pseudohypoparathyroidism. The USMLE will give you type I, which presents with what is referred to as the Albright hereditary

	osteodystrophy phenotype (i.e., shortened 4th + 5th metacarpals, round face, and bony abnormalities). Osteoma cutis (nodules of subcutaneous ossification) can be seen (on retired NBME Q). - In pseudohypoparathyroidism type I, there is no response to exogenous PTH, so we will not see any changes in serum calcium, serum phosphate, or urinary cAMP. - Normally, exogenous PTH would cause an increase in serum calcium, a decrease in serum phosphate, and an increase in urinary cAMP. - Patients with pseudohypoparathyroidism will have low serum calcium, high serum phosphate, and high serum PTH levels. That is, the calcium and phosphate levels appear as though the patient has low PTH, when in reality there is just insensitivity to it. PTH goes up due to decreased negative feedback at the calcium-sensing receptors at the parathyroid glands, since serum calcium is low.
Exogenous calcium administration	- After calcium is given, we expect $\uparrow$ serum Ca^{2+}, $\downarrow$ PTH, $\uparrow$ 24,25-D3. - If we inject calcium we induce negative-feedback at the calcium-sensing receptors at the parathyroid glands, so PTH is suppressed. - Since PTH is suppressed, there is $\downarrow$ 1α-hydroxylase activity in the kidney, meaning 1,25-D3 goes $\downarrow$ and 25-D3 builds up. But as mentioned earlier, 25-D3 is immediately shunted to an inactive storage form called 24,25-D3, so the latter $\uparrow$.
Milk-alkali syndrome	- Diagnosis of exclusion on USMLE, meaning we eliminate to get there. - Textbook scenario is hypercalcemia and metabolic alkalosis (i.e., high bicarb) in someone taking too many calcium carbonate antacid tablets for GERD or dyspepsia (indigestion/upset stomach). This can lead to urolithiasis. - What the USMLE vignette will do, however, is not mention the patient's Hx, but will simply give you a kidney stone + hypercalcemia + high bicarb (or toward the upper limit of normal). Then you eliminate to get there, where you say, "Even though they don't mention the textbook presentation, no other answers explain the high calcium + high bicarb." - Normal bicarb is 22-28 mEq/L, but an NBME Q for milk-alkali syndrome gives a bicarb of 27, where this throw students off. Acid-base disorders can occasionally have bicarb in the normal range, albeit in the direction you expect.
Space travel (not a troll)	- The USMLE asks a Q where an astronaut goes off into space for X number of months + eventually comes back to earth. They ask, compared to before going to space, what would be expected now for the following variables. Answer is: - $\downarrow$ Urinary cAMP, $\downarrow$ PTH, $\downarrow$ bone density. - In space, bone density $\downarrow$ due to $\downarrow$ weightbearing from $\downarrow$ effects of gravity (similar to osteoporosis mechanism). This will cause $\uparrow$ calcium leaching out of the bone, which could theoretically suppress PTH. - Since PTH normally $\uparrow$ urinary cAMP, if PTH is $\downarrow$, urinary cAMP is also $\downarrow$. - The question doesn't ask about serum calcium levels, but in theory they are probably within the normal range (albeit $\uparrow$ compared to baseline). This is the same for osteoporosis, <i>where serum calcium levels are normal</i>.
Osteoporosis	- Serum calcium, phosphate, PTH, and ALP are all normal, as mentioned above. - Including it in this table because of its yieldness. Students often erroneously think calcium is low. It's not. Old granny who has reduced bone density will have normal labs. Same with osteoporosis due to exogenous glucocorticoids. - But USMLE does want $\downarrow$ osteoblast activity, $\uparrow$ osteoclast activity, and $\uparrow$ RANK-L activity in osteoporosis.

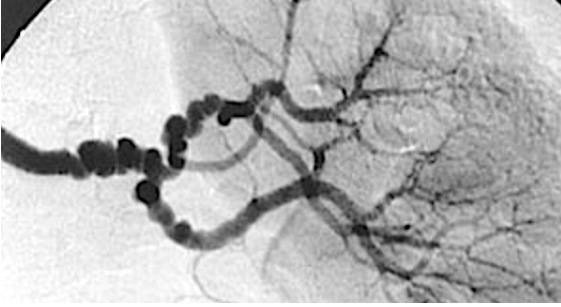
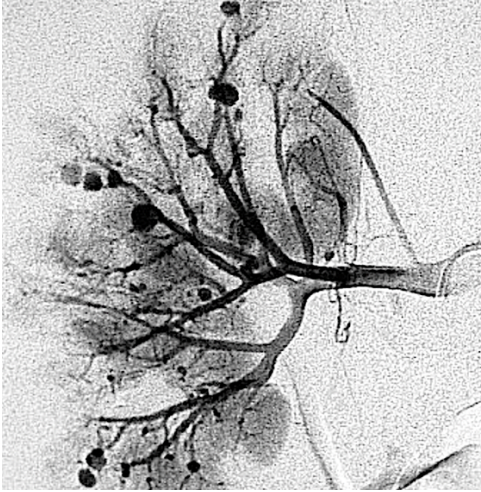


RAAS / Adrenal hormone basics	
RAAS	- Renin-angiotensin-aldosterone system - ↓ Renal perfusion stimulates renin secretion from juxtaglomerular cells (JGC) of the PCT of the kidney. JGCs are modified smooth muscle cells that sense ↓ perfusion. - Students will often tell me "low BP" is the impetus for renin secretion. This is not true. A patient can have high BP and reduced renal perfusion. - The liver produces a protein called angiotensinogen that is cleaved by renin in plasma into angiotensin I. - Angiotensin I goes to the lungs, where angiotensin-converting enzyme (ACE) cleaves angiotensin I into angiotensin II (AT II). - Angiotensin II has multiple roles, but one of them is to go to the zona glomerulosa of the adrenal cortex (most superficial hormone-producing layer) to upregulate aldosterone synthase, thereby ↑ aldosterone. - In other words, ↓ renal perfusion → ↑ AT II + aldosterone. - If a patient's aldosterone level is high, we look at renin (i.e., ↑ or ↓) to help us ascertain the cause of the ↑ aldosterone. - If renin and aldosterone are both ↑, we know the cause is ↓ renal perfusion. We call this renovascular hypertension, where we either have renal artery stenosis (RAS) caused by atherosclerosis (older patient with CVD risk factors) or fibromuscular dysplasia (FMD; vascular smooth muscle proliferation disorder in young women). "Reninoma" is possible but nonexistent on USMLE. - If aldosterone is ↑ but renin ↓, we know the cause of the ↑ aldosterone is primary adrenal, which means Conn syndrome (aldosterone-secreting tumor) or zona glomerulosa hyperplasia (secreting aldosterone). The ↑ aldosterone causes ↑ plasma volume, which ↑ renal blood flow, thereby ↓ renin through negative-feedback. So in short: - ↑ Aldosterone + ↑ renin: Renovascular hypertension (RAS or FMD). - ↑ Aldosterone + ↓ renin: Primary adrenal (Conn or hyperplasia).
Aldosterone	- Produced in zona glomerulosa of adrenal cortex. - Causes ↑ plasma Na^+, ↓ K^+, ↑ HCO_3^-, ↑ pH. - In other words, all of the arrows go the same direction as aldosterone except for K^+. - Aldosterone goes to the basolateral membrane (side of blood) of the cortical collecting duct in the kidney to ↑ Na^+/K^+ ATPase pumps. For every 1 ATP utilized, we get 2 K^+ secreted

	from the blood into the tubular cell (which ultimately leaves the body) and 3 Na⁺ reabsorbed from the tubular cell into the blood. - When Na⁺ is pulled out of the tubular cell into the blood, this favors a high-low gradient from the urine into the cell for sodium, causing ENaC (a sodium channel) on the apical membrane to indirectly ↑ in activity, which pulls Na⁺ out of the urine. - In addition, aldosterone upregulates an H⁺ ATPase on the apical membrane (side of urine) of the cortical collecting duct, causing ↑ serum HCO₃⁻ (i.e., ↓ plasma H⁺ to mop up HCO₃⁻, so HCO₃⁻ ↑).  1) Na⁺/K⁺-ATPase activation by aldosterone. 2) This creates a high-low gradient for Na⁺ from the urine into the cell. 3) ENaC then upregulates to absorb Na⁺ 4) K⁺ and H⁺ are just secreted (excreted through the walls of the tubular cell). - The main purpose of aldosterone is to regulate fluid status in the body. Water follows Na⁺, so the strong Na⁺ reabsorptive effect functions to retain water. - The interesting point to note is that although ADH (vasopressin) ↑ free-water reabsorption (without pulling Na⁺ in as well), this effect is more to regulate tonicity of the blood (i.e., whether plasma Na⁺ is ↑ or ↓) by “diluting” out serum Na⁺. Vasopressin still has some effect on volume status, but aldosterone’s effect is more robust.
AT II	- In addition to upregulating aldosterone synthase in the zona glomerulosa of the adrenal cortex, AT II has 3 other very important roles for USMLE: ○ 1) Constricts the efferent arterioles leaving the kidney. ▪ This causes ↑ hydrostatic pressure backup at the glomerulus, which maintains GFR in the setting of ↓ renal perfusion. This means ↑ FF, since the latter = GFR/renal perfusion. ▪ It makes sense that AT II constricts the efferent arterioles, because when renal perfusion is ↓, GFR will ↓ unless we get a “squeezing of the hose” / backup effect leaving the kidney, which functions to maintain the same glomerular filtration in spite of that ↓ in renal blood flow. ○ 2) Constricts peripheral arterioles. ▪ Helps to maintain basal blood pressure. ▪ Main reason this function of AT II is HY for USMLE is because they want you to know that ACE inhibitors (e.g., lisinopril) or AT II receptor blockers (e.g., valsartan) ↓ afterload on the heart by relieving the constrictive effects of AT II on peripheral arterioles. This promotes ↑ ejection fraction, since there is now less force for the heart to pump against. This effect at helping to ↑ EF, in addition to their ability to ↓ cardiac remodeling, is why ACEi/ARBs are first-line meds in heart failure. ○ 3) Promotes Na⁺ reabsorption in the PCT of the kidney. ▪ Contributes to fluid retention in the setting of ↓ renal perfusion without the need for aldosterone to act distally. ▪ For example, in pre-renal azotemia, FeNa is ↓ due to the ↑ reabsorption of Na⁺ proximally by AT II. If you’re confused by this, I discuss all of this stuff in detail in the HY Renal PDF.
Cortisol	- Produced in zona fasciculata of adrenal cortex. - Three most important roles for USMLE:

	○ 1) Maintains basal BP by $\uparrow$ α1-receptor expression on peripheral arterioles. Norepinephrine and epinephrine (catecholamines produced in adrenal medulla) then bind to α1-receptors, causing arteriolar constriction and $\uparrow$ BP. ○ 2) Maintains basal blood glucose levels by $\uparrow$ insulin resistance at peripheral tissues (physiologic/healthy if cortisol levels are normal) and $\uparrow$ gluconeogenesis. ○ 3) For repro/obgyn, it also accelerates fetal lung maturity by $\uparrow$ surfactant production by the lamellar bodies within type II pneumocytes. - ACTH causes cortisol secretion by the zona fasciculata of the adrenal cortex. - Cortisol will then induce negative-feedback at the hypothalamus and anterior pituitary to $\downarrow$ CRH and ACTH, respectively. - If cortisol is $\downarrow$, CRH and ACTH go $\uparrow$ due to lack of negative-feedback. - The precursor to ACTH is proopiomelanocortin (POMC), which is cleaved into both ACTH as well as α-MSH (melanocyte-stimulating hormone). - $\downarrow$ cortisol causing $\uparrow$ CRH $\rightarrow$ $\uparrow$ POMC $\rightarrow$ $\uparrow$ ACTH + $\uparrow$ α-MSH is why hyperpigmentation is seen in Addison disease.
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Renovascular hypertension	
Renal artery stenosis (RAS)	- Narrowing of one or both renal arteries due to atherosclerosis that causes $\uparrow$ renin-angiotensin-aldosterone system (RAAS) and $\uparrow$ BP. - Q will be patient over the age of 50 with cardiovascular disease risk factors, such as diabetes, HTN, and/or smoking. - Patients who have pre-existing HTN causing atherosclerosis leading to RAS will often have 10-20 years of background HTN that then <i>becomes accelerated</i> over a few-month to 2-year period. What this means is: the slowly developing atherosclerosis in the renal arteries finally reaches a point at which the kidney is unable to maintain autoregulation, and the RAS is now clinical (i.e., accelerated HTN of $\uparrow\uparrow$ BP within, e.g., 3 months). - Another way USMLE will give RAS is by giving $\uparrow$ BP in patient with significant evidence of atherosclerotic disease (i.e., Hx of coronary artery bypass grafting, intermittent claudication), and then ask for the most likely cause $\rightarrow$ RAS. You have to say, "Well he clearly has atherosclerosis in his coronaries and aortoiliac vessels, so that means he'll have it in the renal arteries too." - Q can say older patient with carotid bruit has recent $\uparrow\uparrow$ in BP and then ask for diagnosis $\rightarrow$ answer = RAS. Similar to above, if the patient has atherosclerosis in one location (i.e., the carotids), then he/she will have it elsewhere too. - HY factoid about RAS is that ACEi or ARBs will cause renin and/or creatinine to go up. This is a HY point that is often overlooked and is asked on NBMEs. I have not seen NBME care whether it's uni- or bilateral in this case. $\rightarrow$ Kidney can autoregulate across flux in perfusion. Patients with already-compromised renal blood flow are more sensitive to the subtle $\downarrow$ in filtration fraction that occurs secondary to ACEi/ARB use, so renin/creatinine $\uparrow$. - If USMLE gives you unilateral RAS, renin is only $\uparrow$ from that kidney. The other kidney will not produce $\uparrow$ renin. - After renin and aldosterone levels are obtained, MR angiography of the renal vessels is what USMLE wants for the next best step in diagnosis.
Fibromuscular dysplasia (FMD)	- The answer on USMLE for narrowing of the renal arteries in a woman 20s-40s. - Not the same as renal artery stenosis, and not caused by atherosclerosis. - If you broadly say "renal artery stenosis," that specifically refers to atherosclerosis of the renal arteries in patient >50 with CVD. - FMD is tunica media hyperplasia (not dysplasia, despite the name) that results in a "string of beads" appearance on renal angiogram.

	 - MR angiography is answer on NBME for diagnostic modality. - Can affect the carotid vessels. A 2CK Surg Q gives FMD vignette and also says there is 25% occlusion of one of the carotids.
Polyarteritis nodosa (PAN)	- As discussed in the cardio PDF, this is a medium-vessel vasculitis that causes a “string of pearls” appearance of the renal vessels. This can be confused with FMD, but note the difference in the imaging. FMD shows more beading along the larger/proximal arterial sections, whereas PAN shows pearls more at the vascular termini, with the more proximal parts more likely to be spared.  Angiogram of PAN shows pearls that are more distal in the renal vasculature; FMD, in contrast, shows more proximal beading, with the termini not as conspicuously involved. - There is an NBME Q where they list both PAN and FMD as answer choices and it relies on you knowing the angiogram to get it right. They don't mention hepatitis B, but answer is PAN. That's why I'm harping on this here. - Causes fibrinoid necrosis. - Can be caused by hepatitis B. - Spares the pulmonary vasculature.
Renal artery thrombosis	- The answer on USMLE for hypertension in a neonate following umbilical artery catheterization. USMLE simply wants “↑ renin, ↑ aldosterone” as the answer. - Sounds weird, but you need to know umbilical artery catheterization is a major risk factor for renal artery thrombosis in neonates. - 2CK NBME Q gives brief umbilical artery catheter insertion in kid born 26 weeks' gestation in order to monitor blood pressure for a pneumonia. 3 weeks later, he has ↑ BP (128/86) → answer = “↑ renin, ↑ aldosterone” as answer. - BP in term neonates should be ~60/40. In a preemie 29 weeks' gestation, it should be ~50/30.

Primary adrenal Ddx affecting aldosterone	
Conn syndrome	- Aldosterone-secreting tumor of the zona glomerulosa of the adrenal cortex. - Classically causes $\uparrow \text{Na}^+$, $\downarrow \text{K}^+$, $\uparrow \text{HCO}_3^-$, $\uparrow \text{pH}$. - Renin is $\downarrow$ due to negative-feedback at the kidney. In other words, the $\uparrow$ aldosterone from the tumor $\uparrow$ fluid retention, which causes $\uparrow$ renal blood flow, which shuts off renin production. - In ~50% of aldosterone-derangement Qs, Na^+ will be normal. This confuses students, but it's because the body has good ability to regulate Na^+ (i.e., also via ADH). - Tx = spironolactone (aldosterone receptor antagonist) prior to surgery.
Familial hyperaldosteronism	- There are gene mutations that can cause $\uparrow$ primary adrenal production of aldo. - If the USMLE gives you Q where various family members have $\uparrow$ BP + $\uparrow$ aldosterone and they don't tell you any other info, including the renin level, the answer is primary hyperaldosteronism, not renovascular hypertension (i.e., FMD or RAS). - There is a Q on an NBME form where they do this, and the student says, "But wait, how are we supposed to know it's primary and not FMD/RAS?" It's because if $\uparrow$ aldo is familial, it's likely due to gene mutations causing $\uparrow$ primary production from the zona glomerulosa.
Addison disease	- Autoimmune destruction of the adrenal cortex, leading to $\downarrow$ aldosterone <i>and</i> $\downarrow$ cortisol. - I haven't seen the USMLE ever care about "decreased androgens," so the effect is predominantly on the zona glomerulosa and fasciculata. - Called "primary adrenal insufficiency" because the adrenal gland itself is under-producing hormones. - Addison is idiopathic / can be seen in patients with Hx of other autoimmune diseases, either in themselves or family members (i.e., the vignette throws in the side-detail that the brother has RA, or the sister has IBD, or the patient has Hx of thyroiditis). - $\downarrow \text{Na}^+$, $\uparrow \text{K}^+$, $\downarrow \text{pH}$, $\downarrow$ bicarb. - Blood pressure will either be low-normal or low, due to both $\downarrow$ aldosterone as well as $\downarrow$ cortisol. I explicate this because students tend to just mention the $\downarrow$ aldosterone as the reason BP is $\downarrow$, but the effect of $\downarrow$ cortisol is HY for USMLE. - ACTH and α-MSH are both $\uparrow$ due to $\downarrow$ cortisol $\rightarrow$ hyperpigmentation. Although I'd say they only mention this in ~50% of vignettes. - Patients can present with eosinophilia (high eosinophils). This is because cortisol normally plays a role in helping to sequester eosinophils within the spleen + facilitates eosinophil apoptosis, so $\downarrow$ cortisol means $\uparrow$ eosinophils. Don't go chasing stool ova and parasites. Qs can give you eosinophils of 15-25%, where you're like wtf? They should normally be <5%. - $\downarrow$ cortisol causes chronic fatigue syndrome. This is one of the most over-looked details in NBME Qs and in real life. Any time a patient has ongoing chronic fatigue that is unexplained, cortisol levels should be checked. If they are $\downarrow$, hydrocortisone will immediately boost the patient. - Classic Addison vignette is patient (young or old, I've seen both) who has ongoing fatigue + hyperpigmentation, and then the answer is just "ACTH stimulation test." - The ACTH stimulation test (cosyntropin test) is used to diagnose adrenal insufficiency. Shows up on NBME. Exogenous ACTH should normally cause a robust $\uparrow\uparrow$ in cortisol from the adrenal fasciculata. In the setting of primary adrenal insufficiency, however, since the adrenal gland is fucked up, giving exogenous ACTH won't do jack shit. - Tx for Addison = fludrocortisone +/- hydrocortisone. Fludrocortisone is a mineralocorticoid similar to aldosterone. Hydrocortisone is a glucocorticoid similar to cortisol.
Waterhouse-Friderichsen	- Meningococcal septicemia causing bilateral hemorrhagic necrosis of the adrenal glands, resulting in $\downarrow$ aldosterone and $\downarrow$ cortisol.

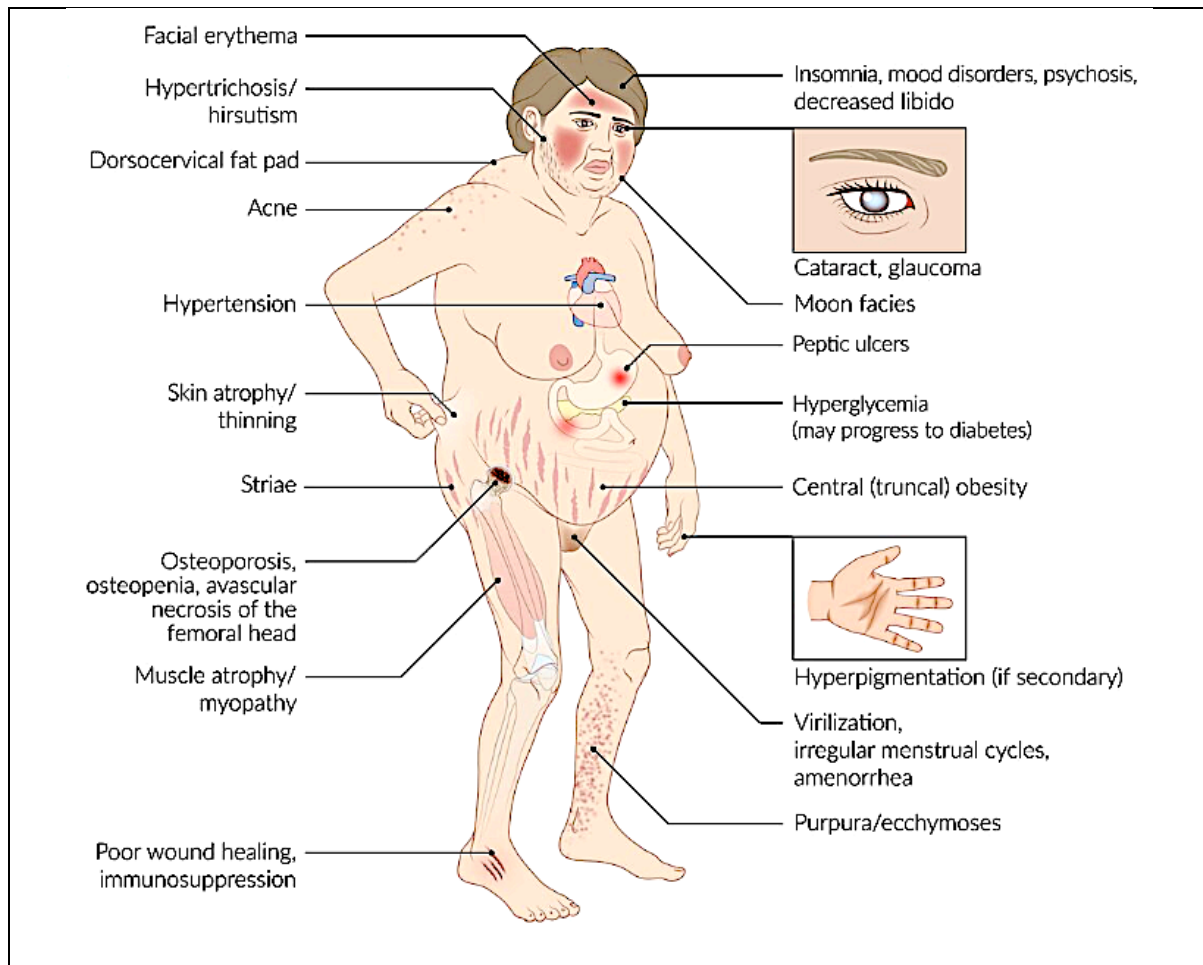
	- Vignette will be ↓ BP + ↓ Na ⁺ + ↑ K ⁺ + ↓ bicarb in patient with meningococcal infection (e.g., recent meningitis + non-blanching rash), where answer = ACTH stimulation test.
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Adrenal crisis

- Patients who are on chronic glucocorticoids (i.e., prednisone) for autoimmune conditions such as IBD, RA, SLE, etc., will have a chronically suppressed adrenal gland that is atrophied, since prednisone is a cortisol analogue that induces negative-feedback and shuts off ACTH production.
- This means that if these patients encounter a stressor such as surgery, trauma, infection, or thyroid storm, where glucocorticoids normally experience ↑ rate of consumption, they are unable to “mount a stress response,” (i.e., unable to produce an acute ↑ in cortisol), so they are prone to acute glucocorticoid deficiency and a precipitous ↓↓ in BP.
- Q will tell you patient has SLE treated with prednisone who goes into surgery and gets a ↓↓ drop in BP to 80/40 + fluids and pressors don’t help; Q asks next best step → answer = IV hydrocortisone or methylprednisolone. What we are doing is replenishing the glucocorticoid. “Pressors” can refer to NE, E, or even desmopressin.
- Remember that without cortisol (or cortisol equivalent, i.e., exogenous glucocorticoid), we can’t get adequate α₁-receptor expression on arterioles, so even if NE and E are floating around in normal amounts (or given exogenously), they can’t bind to α₁-receptors to maintain BP.
- This concept is referred to as: “glucocorticoids are permissive of the effects of catecholamines,” where they literally “permit” NE and E to do their job.
- The Q can also tell you patient being treated with prednisone for SLE or RA experiences an acute thyroiditis (remember autoimmune diseases go together) and gets ↓↓ in BP → answer = hydrocortisone (after fluids). Or patient with autoimmune disease gets an infection + ↓↓ in BP → answer = hydrocortisone (after fluids).
- I reiterate this concept because it is exceedingly HY on USMLE and is missed by students all of the time.
- Self-explanatory, but similar to Addison disease or Waterhouse-Friderichsen syndrome, if the Q mentions laboratory studies in adrenal crisis, they will be ↓ Na⁺, ↑ K⁺, ↓ pH, ↓ bicarb, and ↑ eosinophils.

Cushing syndrome vs disease

- **Cushing syndrome** = how the patient looks/presents; can refer to any cause of Cushingoid presentation (i.e., exogenous glucocorticoids, ACTH-secreting tumor of anterior pituitary, cortisol-secreting tumor of adrenal cortex, small cell bronchogenic carcinoma secreting ACTH).
- **Cushing disease** = only ACTH-secreting tumor of anterior pituitary.
- If patient walks through the door and appears Cushingoid due to an ACTH-secreting tumor of the anterior pituitary, we would say, “The patient has Cushing syndrome caused by Cushing disease.”
- Highest yield findings for Cushingoid appearance on USMLE, as per my observation, are: **purple striae** on the abdomen / stretch marks, **osteoporosis** (compression fractures of vertebrae), and **avascular necrosis of the femoral head** (hip pain).
- Findings such as buffalo hump, moon facies, central obesity, peripheral muscle wasting, HTN, and diabetes type II are important and pass-level, but you really need to know those 3 bold findings I write above.
- In Peds, Cushings can present as precocious puberty without all of the classic findings.
- Chronically elevated cortisol can cause **hypokalemia** similar to high aldosterone. Sometimes NBME will give you a vignette of Cushings + mention potassium is 3.0 mEq/L (NR 3.5-5.0), and you’re like, “why the fuck is the potassium low. Is aldosterone high?” No. All glucocorticoids have some degree of mineralocorticoid effect at the kidney similar to aldosterone. In patients who have high cortisol, the effect longer term can occasionally cause hypokalemia. This detail is particularly important for 2CK vignettes.



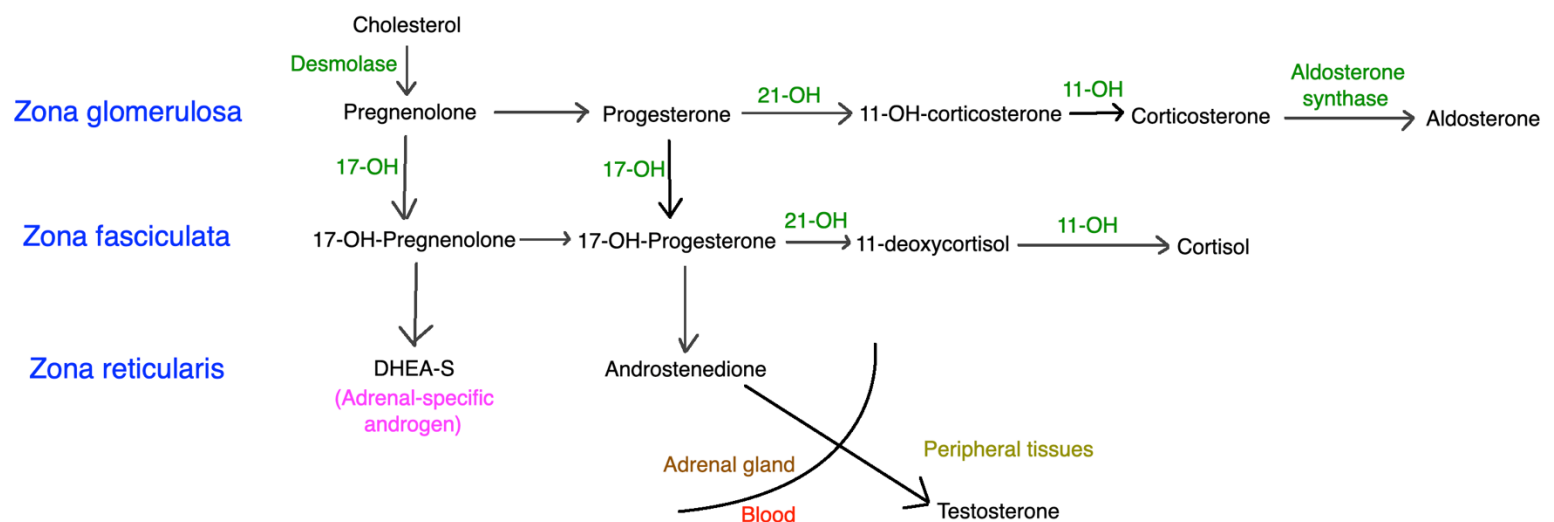
Dexamethasone suppression test

- Dexamethasone is a cortisol analogue that is capable of exerting negative-feedback at the hypothalamus and anterior pituitary. Given to healthy individuals, the result is suppression of ACTH and cortisol levels.
- The dexamethasone suppression test is done to determine the etiology of *endogenous* Cushing syndrome (i.e., the patient isn't taking exogenous steroids). This is because if the patient is taking steroids, clearly we know the cause of the Cushings already.
- Low-dose dex will suppress ACTH and cortisol in healthy patients. It is effectively a "yes or no" test for Cushing syndrome, where if there's suppression of cortisol with low-dose dex, then we can say, "Patient doesn't have Cushings"; if cortisol doesn't suppress, then we know the patient has Cushings, but we just don't know the cause yet.
- High-dose dex helps us differentiate Cushing disease from other causes of Cushing syndrome. This is because only Cushing disease will have suppressed cortisol levels.
- Therefore, if cortisol goes ↓ with high-dose dex, we know the diagnosis is Cushing disease.
- If cortisol does not go ↓ with high-dose dex, we look at ACTH level.
- If ACTH is ↓, we say, the only way that it's possible to have ↑ cortisol with ↓ ACTH is if the diagnosis is an adrenal cortical adenoma of the zona fasciculata secreting cortisol.
- If ACTH is ↑, we know the cause is ectopic ACTH secretion from small cell bronchogenic carcinoma. The reason we know the ACTH isn't from the pituitary is because high-dose Dex didn't suppress cortisol.

Cushing disease	- ↑ ACTH + ↑ cortisol. - Suppresses with high-dose dex. - If high-dose dex is given and they ask for the arrows for how ACTH and cortisol <i>will change</i>, the answer is "↓ ACTH, ↓ cortisol."
Small cell lung cancer	- ↑ ACTH + ↑ cortisol.

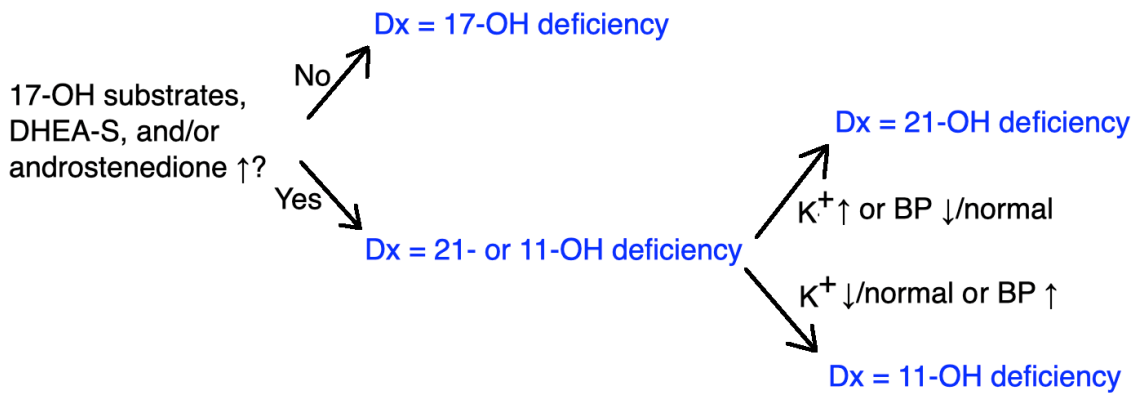
	- Does not suppress with high-dose dex. - If high-dose dex is given and they ask for the arrows for how ACTH and cortisol <i>will change</i>, the answer is “no change ACTH, no change cortisol.”
Adrenal adenoma	- ↓ ACTH + ↑ cortisol. - Does not suppress with high-dose dex. - Same as with small cell lung cancer, if high-dose dex is given and they ask for the arrows for how ACTH and cortisol <i>will change</i>, the answer is “no change ACTH, no change cortisol.”
Exogenous glucocorticoids	- ↓ ACTH + ↓ cortisol (holy shit). - Exogenous glucocorticoids (i.e., prednisone, hydrocortisone) are not the same thing as cortisol. The result is suppression of endogenous cortisol. - Students get this wrong all of the time.

Miscellaneous	
Glucocorticoid psychosis	- Glucocorticoids can cause “glucocorticoid psychosis,” which can present as an overt psychosis, delirium-like presentation, or even just as general mood disturbance, i.e., depression. - The vignette can give you Cushing syndrome + depression, or recent IV methylprednisolone + delirium-like presentation 24-48 hours later.
Licorice	- Licorice contains glycyrrhizic acid, which has mineralocorticoid effect at the kidney similar to aldosterone. - Can cause hypokalemia in those who drink excessive licorice tea.



Congenital adrenal hyperplasia (CAH)	
- Autosomal recessive. - 21-, 11-, and 17-hydroxylase deficiencies. 21-hydroxylase deficiency is most common. - ACTH is ↑ in all three types. This is the mechanism USMLE wants for the hyperplasia. That is, cortisol production is impaired in all three, so ACTH goes ↑ to compensate → stimulates hyperplasia of adrenals. - The USMLE will give a vignette of CAH, followed by “which of the following is the most likely explanation for these findings?” → answer = “Increased ACTH secretion.” Sounds weird, but it’s on NBME. - The relevance of these conditions is that they are a cause of ambiguous genitalia and/or virilization. - Presentations are variable, with Qs giving vignettes in neonates, children, and adults. So the way to differentiate them comes down to electrolytes and BP. - For both 21- AND 11-hydroxylase deficiencies:	

- ↑ 17-OH substrates (i.e., 17-OH-progesterone and 17-OH-pregnenolone).
- ↑ DHEA-S, ↑ androstenedione.
- If you see any of these up in the Q, you know right away answer is either 21- or 11-deficiency. This is because it's impossible to get 17-OH substrates in 17-hydroxylase deficiency.



21-hydroxylase deficiency	- ↑ Serum K⁺, ↓ Na⁺, ↓ pH, ↓ HCO₃⁻, ↓ glucose. - ↓/↔ BP. - Cortisol and aldosterone, as well as their precursors, are all low.
11-hydroxylase deficiency	- ↓/↔ Serum K⁺, ↑/↔ Na⁺, ↑/↔ pH, ↑/↔ HCO₃⁻, ↑/↔ glucose. - ↑ BP. - Cortisol and aldosterone are low, but their precursors, which have some effect, are high.
17-hydroxylase deficiency	- ↓ 17-OH substrates (i.e., 17-OH-progesterone and 17-OH-pregnenolone). - ↓ DHEA-S, ↓ androstenedione. - ↓ Serum K⁺, ↑ Na⁺, ↑ pH, ↑ HCO₃⁻ (high aldosterone). - ↓ glucose (cortisol and its precursor are low).

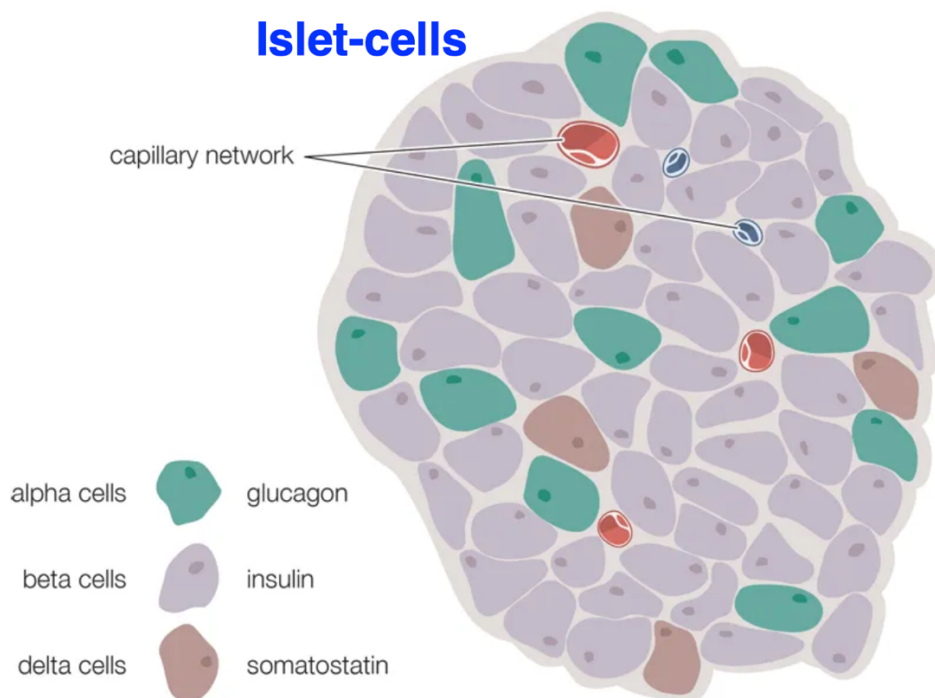
Catecholamine-secreting tumors	
Pheochromocytoma	- Catecholamine-secreting tumor of adrenal medulla. - Derived from neural crest (i.e., chromaffin cells of adrenal medulla). - Occurs in adults; can calcify. - Causes paroxysmal (i.e., comes and goes) ↑↑ BP, which presents as periodic palpitations and pounding headaches. - Because the HTN is paroxysmal, the patient can have BP measured in the office at 120/80, so do not exclude pheo in this case. - Stressors (e.g., sitting up on the examination table) can trigger ↑ NE + E release from the tumor, where BP shoots up to >180/100. This is different from whitecoat hypertension, where young patient has 140/90 measured due to being slightly nervous. - Diagnose with urinary/serum metanephrines. If positive, then do CT abdomen. - Tx = phenoxybenzamine (irreversible α1-blocker). If you give beta-blocker first, this will kill the patient. This is because if you block beta receptors, the NE and E have little to bind to now except α receptors (mainly α1), so we get ↑↑ arteriolar constriction and BP shoots up. This is called "unopposed alpha." - Some students ask about giving propranolol after phenoxybenzamine. The answer is, yes, in theory this can be done. But USMLE doesn't assess this. They want straight-up alpha-blockade. - Pheo can be idiopathic or part of condition such as MEN 2A/B or NF1. - If a patient has medullary thyroid cancer, you must check for pheochromocytoma before doing thyroidectomy. This is because if the patient has MEN 2 and you don't

	check for pheo first, doing surgery will cause ↑↑ BP due to catecholamine release and kill the patient.
Neuroblastoma	- Essentially the “pediatric pheo”; <i>N-myc</i> gene. - Secretes catecholamines; can calcify (same as pheo in adults). - Grows from the median (midline) sympathetic chain. Usually intra-abdominal but can sometimes be in the posterior mediastinum. Sounds odd, but the latter shows up on a Peds CMS form. - Causes ↑ BP and findings such as opsoclonus-myoclonus syndrome (dancing eyes) and violaceous eyelids (sounds like heliotrope rash of dermatomyositis but unrelated; caused by orbital metastases). - Diagnose with urinary homovanillic acid (HVA) and vanillylmandelic acid (VMA), which are metabolites of catecholamines. - Metaiodobenzylguanidine scan can be done after HVA and VMA detected.

Carcinoid tumor	
	- Neuroendocrine tumor of small bowel, appendix, or lung that is S-100 (+) and consists of small blue cells. - Secretes serotonin. - Presents as flushing, tachycardia, diaphoresis, and diarrhea. - Tricuspid regurg common (holosystolic murmur that increases with inspiration) due to tricuspid lesions. - Diagnose with urinary 5-hydroxyindole acetic acid (5-HIAA). - Don't confuse with serotonin syndrome, which is a drug interaction (i.e., starting a monoamine oxidase inhibitor too soon after discontinuing an SSRI, or if a patient takes St John wort with an SSRI).

Polyglandular syndromes	
- All MEN syndromes and Carney complex are autosomal dominant.	
MEN 1	- <i>MEN1</i> gene mutation. - Parathyroid adenoma (or hyperplasia), pituitary adenoma, pancreatic adenoma. - o Zollinger-Ellison syndrome (gastrinoma) is highest yield pancreatic tumor in MEN 1. Will present as recurrent peptic ulcers. I discuss this stuff in detail in the HY Gastro PDF. - o Parathyroid adenoma often presents with calcium urolithiasis, where they will mention flank or groin pain in patient with Hx of ulcers (Zollinger-Ellison). - o Prolactinoma is highest yield pituitary tumor. - o USMLE need not give the full constellation of findings for the MEN syndromes. There is an NBME Q floating around where the patient has a parathyroid adenoma – nothing more – and the answer is just <i>MEN1</i> as the gene that's mutated.
MEN 2A	- <i>RET</i> proto-oncogene mutation. - Parathyroid adenoma (or hyperplasia), pheochromocytoma, medullary thyroid carcinoma. - o <i>RET</i> mutations need not cause the full constellation of findings. NBME Qs can give isolated medullary thyroid carcinoma in a family, and the answer is just <i>RET</i>. - o In theory, isolated parathyroid adenoma could also be <i>RET</i>, but with the NBME example I mentioned above they listed <i>MEN1</i> without also listing <i>RET</i>.
MEN 2B	- <i>RET</i> proto-oncogene mutation. - Marfanoid body habitus, mucosal neuromas, pheochromocytoma, medullary thyroid carcinoma. - o –“Oid” means looks like but ain't. So the patient might look like he/she has Marfan's (i.e., tall, lanky) but doesn't, because this isn't <i>FBN1/2</i> mutation. - o As mentioned earlier, before doing thyroidectomy for medullary thyroid carcinoma, you must check for pheochromocytoma (because stress of surgery

	will ↑↑ catecholamines from the tumor and kill the patient) by first doing urinary metanephrines. If positive, then do CT of the abdomen.
Carney complex	- Hyperpigmentation, atrial myxoma, endocrine hypersecretion. - o The hyperpigmentation can present as perioral melanosis (i.e., hyperpigmentation around the lips/mouth) that sounds similar to Peutz-Jegher syndrome, but instead of the Q focusing on hamartomatous colonic polyps, they mention a cardiac tumor and another endocrine organ hyper-secreting. And you're like huh? - o Atrial myxoma normally shows up in adults; cardiac rhabdomyoma normally shows up in children with tuberous sclerosis. In Carney, the Q can say atrial myxoma <i>in a kid</i>. So it's the only example on USMLE that breaks the rule. - o Endocrine hypersecretion is classically Cushing syndrome (granular hyperpigmentation of the zona fasciculata) or hyperthyroidism. - o The reason you likely haven't heard of Carney complex before is because students coming out of the USMLE didn't even realize this is what they saw.



Islet-cell HY points	
Insulin function	- ↓ Blood glucose by promoting ↑ uptake into cells, especially skeletal muscle and adipose tissue, via induction of GLUT4 translocation to cell surface. ↑ Uptake into the liver occurs via upregulation of glucokinase (hexokinase variant in the liver). - ↑ Glycogenesis by ↑ glycogen synthase activity. - ↑ Lipogenesis by promoting the conversion of excess glucose into fatty acids, followed by storage of the fatty acids in adipose tissue. - ↑ Protein synthesis by enhancing amino acid uptake into cells and promoting ribosome formation. - Inhibition of gluconeogenesis.
Insulin secretion	- Pancreatic beta cells take up glucose via GLUT2 → ATP is produced within the cell → closure of the ATP-gated K⁺ channel → K⁺ builds up within the β-islet cell → cell depolarizes → Ca²⁺ influxes into the cell → this triggers release of insulin-filled vesicles from the β cell.

Insulin binding	- Insulin binds to a tyrosine kinase on the cell surface of skeletal muscle and adipose tissue → this ultimately results in ↑ GLUT-4 expression on the cell surface, which allows for glucose to enter the cell. - USMLE wants you to know that insulin causes: ↑ tyrosine-kinase activity, ↑ serine-threonine kinase activity, and ↓ ubiquitin-mediated proteolysis. - Ubiquitin is normally tagged to proteins for degradation during catabolic processes, but insulin promotes anabolic processes (i.e., protein synthesis). - I talk about this stuff more in the HY Biochemistry and HY Arrows PDFs.
C-peptide	- During the production of insulin, the precursor protein called proinsulin contains a peptide sequence called C-peptide that is ultimately cleaved off. - This means that endogenous insulin and C-peptide are present in equimolar amounts in the circulation. - If a patient has ↑ insulin, we can look at C-peptide levels to know whether that insulin is endogenously produced or exogenously injected. - If C-peptide levels are ↓, the insulin is injected. - If C-peptide levels are normal/↑, the insulin is endogenously produced. - The tricky part for USMLE is with respect to the workup when C-peptide levels are found to be normal/↑. Some students erroneously say, "Oh that's insulinoma!" Chill. Before you jump on CT of the abdomen looking for insulinoma, USMLE first wants you to check "serum hypoglycemic levels," which refers to medications that can cause insulin secretion (i.e., insulin secretagogues), such as sulfonylureas, meglitinides, GLP-1 analogues, and DPP-4 inhibitors. I discuss these more below. If serum levels of these drugs are negative, then we can do CT of the abdomen looking for insulinoma. - High insulin presents as Whipple triad – i.e., 1) hypoglycemia, 2) symptoms / signs of hypoglycemia (i.e., tachycardia, tremulousness), 3) worsens between meals / gets better with meals.
Glucagon	- ↑ Blood glucose by ↑ gluconeogenesis. - ↑ Glycogenolysis (glycogen breakdown into glucose). - ↑ Lipolysis (breakdown of fatty acids into acetyl-CoAs for ketogenesis). - ↑ Protein catabolism (i.e., ↑ ubiquitination). - USMLE wants you to know glucagon ↑ adenyl cyclase activity and ↑ cAMP. And in theory, β-blockade can reverse the effects of glucagon by ↓ cAMP (i.e., ↓ G-α-s G-protein activity). - Glucagonoma causes a combo of ↑ serum glucose and a body rash known as necrolytic migratory erythema.
Somatostatin	- As mentioned earlier, somatostatin is aka growth hormone-inhibiting hormone (GHIH) and shuts off its secretion from the anterior pituitary. It also ↓ secretion of insulin and glucagon from the pancreas. - Somatostatinoma presents with steatorrhea. - I discuss this stuff in more detail in the HY Gastro PDF.

Diabetes mellitus (DM)

- The word diabetes means polydipsia + polyuria (i.e., drinking a lot and peeing a lot).
- I see students frequently misconstrue diabetes mellitus with diabetes insipidus, since both vignettes will give patients who have increased thirst and urination.
- Three key details that distinguish acute diabetes mellitus from diabetes insipidus is that DM vignettes can often mention weight loss, vomiting, and/or visual changes. Weight loss is also seen in acute type II, even though the patient is usually overweight. Weight loss occurs due to ↑ catabolism. Visual changes occur due to glucose moving into the lens of the eye → conversion via aldose reductase into sorbitol, which has high osmotic pull → water enters lens causing distortion.
- So if vignette says ↑ thirst + ↑ urination +/- weight loss, vomiting, or visual changes, this is DM, not DI.
- For extensive info on the effects of diabetes on different organ systems (i.e., cardio, renal, ophthal, GIT, pregnancy, etc.), use those corresponding HY PDFs I've made. I will keep the focus here on endocrine and electrolytes / acid-base.

- Diagnosis of DM is made by measuring two fasting glucoses >126 mg/dL, any one random fasting glucose >200 mg/dL, or an HbA1c of >6.5%.

Type I	- Absence of insulin; autoimmune-mediated; no one specific gene, although HLA-DR3/4 are associated. - Often described as “genetic susceptibility with environmental trigger” (i.e., viral infection, usually Coxsackie B). - Occurs almost always in children or teenagers of normal BMI. - Pathogenesis is usually combo of an antibody- (anti-GAD65 or -ZnT8) and T-cell-mediated process that leads to destruction of β-islet cells in the tail of the pancreas $\rightarrow$ $\downarrow$ insulin secretion $\rightarrow$ inability to recruit GLUT4 to cell surface of skeletal muscle and adipose tissue $\rightarrow$ inability to take up glucose into cells $\rightarrow$ cells are starved of energy $\rightarrow$ $\uparrow$ gluconeogenesis + ketogenesis. - Hyperglycemia occurs as a result of continued glucose production by the liver (gluconeogenesis) in combination with $\downarrow$ uptake by skeletal muscle and adipose tissue. - Biopsy of pancreas shows inflammatory infiltrate in acute type I DM; in late type I (i.e., years later), it shows atrophy and fibrosis. - It is the absence of insulin in type I DM that enables ketogenesis. In type II DM, ketones are low, not high, because insulin is present. - Ketone bodies (USMLE likes β-hydroxybutyrate) are highly acidic, so blood pH falls. We have high anion-gap metabolic acidosis in DKA (part of MUDPILES). - Vignette will give kid or teenager with polyuria, polydipsia, and sometimes vomiting, who has severe metabolic acidosis (bicarb $\downarrow$), which is due to diabetic ketoacidosis. Breath can be fruity in odor (smell of ketones). - In DKA, we will have: - $\downarrow$ pH, $\downarrow$ HCO_3^-, $\downarrow$ CO_2. - Ketone bodies are acidic $\rightarrow$ causes metabolic acidosis (decreased bicarbonate). - CO_2 is acidic, so we blow that off to compensate. Kussmaul breathing is deep, labored breathing seen in DKA. - $\uparrow$ β-hydroxybutyrate, $\uparrow$ serum osmolality, $\uparrow$ anion gap. - β-hydroxybutyrate is a ketone body. Since we have ketosis (ketone production), it will be increased. I have seen this specific ketone body as an arrow on an offline Step 1 NBME. - Serum osmolality is increased because serum glucose is markedly increased (insulin isn't present to drive glucose into cells). Even though serum sodium is low (so you might think osmolality is low similar to SIADH), the $\uparrow$ glucose makes serum osmolality high. - DKA is the D in MUDPILES, so anion gap is up. - $\downarrow$ Serum Na^+, $\uparrow$ serum K^+, $\downarrow$ total body K^+, $\uparrow$ GFR. - Serum sodium is decreased mostly due to dilutional hyponatremia. As mentioned earlier, serum osmolality is high, which causes fluid retention intravascularly $\rightarrow$ dilutes out serum sodium. In addition, low insulin means less glucose enters cells $\rightarrow$ less ATP production $\rightarrow$ less cellular Na^+/K^+-ATPase activation $\rightarrow$ sodium not pumped out of cells. - We describe the state of K^+ as: hyperkalemia (high serum K^+) despite a low total body potassium. - Serum potassium is high for two main reasons: 1) Potassium-proton shift: Acidosis (due to ketosis) causes protons in the blood to exchange with potassium in the cell via the H^+/K^+-antiporter (i.e., H^+ goes into cell; K^+ moves out). 2) Insulin drives potassium into cells (and we don't have insulin in DKA): Insulin normally leads to the upregulation of a Na^+/K^+-ATPase antiporter that moves 3Na^+ out for every 2K^+ in. If insulin is low, then less K^+ will be moved into the cell $\rightarrow$ hyperkalemia (high potassium in blood). - Total body potassium is low because of increased losses at the kidney. Since serum potassium is high, more potassium is filtered through the glomerulus, so more is excreted (i.e., the kidney thinks there's too much potassium in the body since more is being filtered, so it tries to get rid of more of it). However, this causes
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	bodily depletion of K^+ because the processes that cause hyperkalemia are unmitigated, so the kidney continues to excrete potassium. - GFR is increased because of hyperfiltration secondary to hyperglycemia. That is, since serum glucose is high, more glucose is filtered through the glomerulus, which pulls water with it. - Tx for DKA is normal saline first (i.e., 0.9% NaCl), prior to giving insulin. - Serum potassium will drop as insulin is given. If the Q tells you patient has DKA + shows you normal potassium (even though it's usually high) + insulin is given + patient now has arrhythmia, the answer is potassium for which electrolyte is disturbed (i.e., potassium was normal prior to giving insulin, but now it's been driven into cells and the patient's developed hypokalemia). - Giving bicarb is always a wrong answer for low pH in DKA. - Don't confuse DKA with a hypoglycemic episode. The latter will be a type I diabetic who went out to do exercise or who was cleaning the roof of his house, who is now found unconscious → if insulin dose is not decreased prior to exercise, patient can get hypoglycemia. The Tx is intramuscular glucagon or dextrose in water. Step 1 wants you to know glucagon will increase cAMP. - Sometimes the USMLE can give a bit of a weird Q where they say a patient is given insulin for diabetes but has prolonged hypoglycemia + they ask why → answer = "deficiency of counter-regulatory glucagon." You need to know that in advanced diabetes, the glucagon-producing α-islet cells of the pancreas can also get fucked up. I've also seen this Q asked for chronic alcoholism and pancreatectomy; in both cases, if insulin is given, prolonged hypoglycemia can result from "deficiency of counter-regulatory glucagon."
Type II	- Insulin resistance; seen usually in high-BMI adults, but can rarely show up in Peds (2CK Peds form gives type II in a 17-yr-old). - High BMI causes insulin resistance because adipose tissue secretes adipokines that impair insulin signaling. - Insulin is <i>high</i> initially (hyperinsulinemia) due to β-cell hypertrophy in order to compensate for peripheral insulin resistance. - Ketones are absent/low because the presence of insulin inhibits ketogenesis. - The two above points are the highest yield pieces of info for USMLE. - Biopsy of the pancreas initially shows β-cell hypertrophy, but years later will show amyloid deposition and cell shrinkage. - Same as with type I, hyperglycemia occurs as a result of continued glucose production by the liver (gluconeogenesis) in combination with ↓ uptake by skeletal muscle and adipose tissue. - Non-ketotic hyperosmolar hyperglycemic state (HHS) is seen in type II; DKA is strictly type I on USMLE. - In HHS, glucose is usually >600 mg/dL (in DKA, it need not be this high) and bicarb is usually not below 20 mEq/L (NR 22-28); in DKA, bicarb can be as low as it wants (I've seen 5 mEq/L on NBME). - The USMLE vignette will give massive paragraph Q where they mention glucose is 700 and then ask for the cause of mental status change → answer = "hyperosmolality," or simply "hyperglycemia," or "osmotic diuresis" (causes severe dehydration). I point this out because this is particularly prevalent on 2CK NBMEs, where they'll give a 15-line bullshit paragraph with tons of lab values that the student spends 9 minutes reading, but meanwhile you can see glucose is 650 and then the answer is just "osmotic diuresis" or "hyperosmolality"; should take two seconds to answer. - Mental status change is technically due to hyperosmolality in the ECF of the brain pulling water out of the ICF (i.e., the cells). An NBME Q has "hyperosmolality" as the answer to this. - Treat HHS with normal saline first prior to insulin. - I discuss the various meds for type II diabetes later in this PDF.
Other causes	- Hereditary hemochromatosis ("bronze diabetes") → iron deposition in tail of the pancreas, leading to impaired fasting glucose / overt diabetes. Hyperpigmentation is due to hemosiderin deposition. Patients can also get cardiomyopathy, infertility, and pseudogout.

	- Cushing syndrome (as discussed earlier) can cause diabetes due to the insulin resistance caused by glucocorticoids. - Acromegaly (growth hormone causes insulin resistance). - Alcoholism can cause chronic pancreatitis that not only leads to exocrine pancreatic burnout and steatorrhea, but can also cause diabetes due to loss of islet cells. - Pancreatectomy is self-explanatory. If you remove some of the tail of the pancreas, this can lead to diabetes. - Tacrolimus is a HY drug that can cause diabetes. Other drugs can do this, but tacrolimus is the notable one for USMLE. Tacrolimus is an immunosuppressant that ↓ IL-2 synthesis and T cell stimulation.
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Electrolytes	
Calcium	- 8.4-10.2 mg/dL. - Low Ca causes “up” presentation of tetany and hyperreflexia. - High Ca causes “down” presentation of muscle flaccidity and hyporeflexia. It is also associated with delirium/confusion (hypercalcemic crisis). - Low calcium HY causes are rickets/osteomalacia (↓ vitamin D), secondary hyperparathyroidism (renal failure), post-thyroidectomy (concomitant loss of parathyroids), DiGeorge syndrome (agenesis of 3rd and 4th pharyngeal pouches), loop diuretics (↓ paracellular reabsorption), and hypomagnesemia (hypo-Mg can cause hypo-Ca and hypo-K non-responsive to supplementation; usually seen in alcoholics). - High calcium HY causes are sarcoidosis (due to ↑ vitamin D), primary hyperparathyroidism (adenoma, hyperplasia, MEN 1/2A), metastatic malignancy, multiple myeloma, and thiazides (cause - Ca reabsorption in DCT).
Sodium	- 135-145 mEq/L. - Sodium derangement (high and low) causes CNS dysfunction – i.e., confusion, stupor, or coma. - High Na HY causes are primary hyperaldosteronism (e.g., Conn), renal artery stenosis, fibromuscular dysplasia, dehydration, diabetes insipidus. - Low Na HY causes are Addison disease, Waterhouse-Friderichsen, psychogenic polydipsia, SIADH. - Na can often be normal in USMLE vignettes despite your expectation that it might be characteristically deranged – e.g., patient has Conn syndrome and yet the sodium is normal, and you’re like wtf? (because you expect it to be elevated); this is typical for USMLE vignettes.
Potassium	- 3.5-5.0 mEq/L. - Potassium derangement (high and low) causes cardiac dysfunction → arrhythmia (this contrasts with Na, which causes CNS dysfunction). - High K HY causes are Addison disease, renal failure, polypharmacy with agents such as potassium-sparing diuretics and digoxin. - Low K HY causes are primary hyperaldosteronism, renal artery stenosis, fibromuscular dysplasia, Cushing disease (chronically high glucocorticoid levels can cause hypo-K due to distal renal secretion similar to mineralocorticoids), vomiting, diarrhea, loop diuretics and thiazides, and hypomagnesemia (hypo-Mg can cause hypo-K and hypo-Ca non-responsive to supplementation; usually seen in alcoholics).
Phosphate	- 3.0-4.5 mg/dL. - High phosphorus HY causes are hypoparathyroidism, secondary hyperparathyroidism (renal failure), tumor lysis syndrome, and sarcoidosis (↑ vitamin D). - Low phosphorus HY causes are primary hyperparathyroidism, rickets/osteomalacia, and refeeding syndrome.
Magnesium	- 1.7-2.2 mg/dL. - Magnesium derangement causes effects similar to Ca – i.e., low Mg presents with an “up” state of hyperreflexia and increased muscle tone; high Mg presents with a “down” state of hyporeflexia and muscle flaccidity.

	- Low Mg HY cause is alcoholism (decreased dietary intake); leads to hypocalcemia and hypokalemia non-response to supplementation.
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Acid/base	
(If you want practice with acid/base USMLE Qs and/or are confused by certain things I list in this table, go through my HY Arrows PDF. I discuss this stuff in comprehensive detail).	
- Normal blood pH: 7.35-7.45. - Normal blood HCO₃⁻: 22-28 mEq/L. - Normal blood pCO₂: 33-44 mmHg.	
Metabolic acidosis	- Low pH caused by low HCO ₃ ⁻ ; pCO ₂ will go down to compensate (CO ₂ is acidic).
High anion-gap metabolic acidoses	- MUDPILES is mnemonic for high-anion-gap metabolic acidoses → Methanol, Uremia (renal failure), DKA, Phenformin (weird drug you don't need to worry about), Iron tablets / Isoniazid, Lactic acidosis, Ethylene glycol (antifreeze), Salicylates (aspirin). - If the patient has metabolic acidosis and the anion gap is normal in the Q, you can eliminate the MUDPILES answers and choose an answer like renal tubular acidosis (RTA). - Anion gap is calculated as $\text{Na}^+ - (\text{HCO}_3^- + \text{Cl}^-)$. - Anion gap of 8-12 is normal. So 13 or greater is high. I specifically write "13 or greater" because there is an NBME Q where anion gap comes out to 12, and they want renal tubular acidosis. I talk in detail in the HY Renal PDF about RTA.
Normal anion-gap metabolic acidoses	- Diarrhea, RTA, adrenal insufficiency, and CAH (21-deficiency) are most important.
Metabolic alkalosis	- High pH caused by high HCO₃⁻; pCO₂ will go up to compensate. - Conn syndrome, renal artery stenosis, fibromuscular dysplasia, vomiting, loops and thiazides (promote RAS activation due to volume depletion).
Respiratory acidosis	- Low pH caused by high CO₂; HCO₃⁻ goes up <i>if chronic</i> to compensate (bicarb is basic). - Acute (normal bicarb): hypoventilation due to opioids, benzos, barbiturates, or severe pain (e.g., rib fractures). - Chronic (high bicarb): COPD, obstructive sleep apnea, obesity, ankylosing spondylitis, severe kyphoscoliosis.
Respiratory alkalosis	- High pH caused by low CO₂; HCO₃⁻ goes down <i>if chronic</i> to compensate. - Acute (normal bicarb): asthma attack, panic attack, most pulmonary emboli, altitude sickness (first day at high altitude). - Chronic (low bicarb): living at high altitude, pregnancy (progesterone upregulates respiratory center).
- Should be noted that primary metabolic acid/base derangements are compensated for instantly by changes in respiration (i.e., it's easy to retain or breathe off CO₂ merely by changing respiratory rate), whereas primary respiratory derangements are compensated for slowly because it takes time for the kidney to alter bicarb excretion. - Winter formula is used to calculate predicted pCO₂ based on any change in bicarb, where: ○ Predicted pCO₂ = $(1.5 \times \text{HCO}_3^-) + 8 \pm 2$. - For example, if a patient has DKA and bicarb is 14 mEq/L, we expect pCO₂ to be 27-31 mmHg. If pCO₂ is lower than 27, then the patient has a concurrent respiratory alkalosis; if higher than 31, then the patient has a concurrent respiratory acidosis.	

Salicylate toxicity

- Tinnitus (ear-ringing) is the most common first symptom. The first sign is tachypnea.
- Aspirin initially causes an isolated respiratory alkalosis (within the first 20 minutes) due to direct upregulation of respiratory centers in the brain.
- However, aspirin itself is an acid (salicylic acid), so it will also cause a metabolic acidosis at the same time.
- We call this a mixed metabolic acidosis-respiratory alkalosis after 20 minutes.
- Therefore:
 - o First 20 mins: acute respiratory alkalosis ($\uparrow$ pH, $\leftrightarrow$ HCO_3^- , $\downarrow$ CO_2).
 - o >20 mins: mixed metabolic acidosis-respiratory alkalosis ($\downarrow/\leftrightarrow$ pH, $\downarrow$ HCO_3^- , $\downarrow$ CO_2 , $\uparrow$ anion-gap).
 - o I have only ever seen NBMEs assess the latter. Qbank is not the same as NBME/USMLE.
- The low bicarbonate we see after 20 minutes **is not compensation**. It just coincidentally goes low because aspirin is an acid.
- Anion gap is high. Salicylates are the S in MUDPILES.
- Potassium is always normal in salicylate toxicity. Sodium, chloride, and pH can be variable. There is an NBME Q where they give low Na in salicylate poisoning.

Renal failure key disturbances

- As discussed earlier, renal failure causes secondary hyperparathyroidism, where we have:
 - o $\downarrow$ Ca^{2+} , $\uparrow$ PO_4^{3-} , $\uparrow$ PTH, $\downarrow$ 1,25-D3.
- However, it is also pass-level you know that in renal failure we also have:
 - o $\uparrow$ K^+ , $\downarrow$ HCO_3^- , $\downarrow$ pH, $\uparrow$ anion gap (U in MUDPILES = uremia = renal failure), with Na^+ variable.
 - o The kidney normally secretes potassium distally in the cortical collecting duct under the action of aldosterone. So if the kidney is fucked up, we can't do that, so K^+ is $\uparrow$.
 - o Likewise, we normally have HCO_3^- reabsorbed in the PCT and H^+ secreted in the cortical collecting duct, so if we can't do either of those things, blood HCO_3^- and pH $\downarrow$.

HY Pulmonary acid/base findings

- Acute asthma attack and PE: $\downarrow$ pCO_2 , no change HCO_3^- , $\uparrow$ pH,
 - o The key here is that these conditions cause acute respiratory alkalosis, not acidosis, as I mentioned in above table. Students normally fuck this up and think pCO_2 is high. pCO_2 only goes $\uparrow$ if the patient becomes tired as an asthma attack prolongs. But initially, for both of these conditions, we have low pCO_2 (holy shit), not high. Bicarb is unchanged because it takes a minimum of 12-24 hours for bicarb excretion to be sufficiently modulated by the kidney.
 - o Tangentially, metabolic acid-base disturbances are instantly compensated for by changes in pCO_2 by simply breathing more or less.
- Patients with COPD and obstructive sleep apnea are **chronic CO_2 retainers**, where we have:
 - o $\uparrow$ pCO_2 , $\uparrow$ HCO_3^- , $\downarrow$ pH.
 - o Bicarb goes up to compensate.

Miscellaneous HY acid/base

Vomiting	- $\downarrow$ Serum K^+, $\downarrow$ Cl^-, $\uparrow$ HCO_3^-, $\uparrow$ CO_2, $\uparrow$ pH. - Very high-yield you know that vomiting causes a hypokalemic, hypochloremic, metabolic alkalosis. - Vomitus contains lots of potassium, chloride, and protons. In addition, volume depletion can promote distal renal losses of K^+ and H^+ secondary to RAAS upregulation. - CO_2 is acidic, so we want to retain it to compensate.
Loop/thiazide diuretic abuse	- $\downarrow$ Serum K^+, $\downarrow$ Cl^-, $\uparrow$ HCO_3^-, $\uparrow$ CO_2, $\uparrow$ pH. - Furosemide is a loop diuretic. It will block the apical $2\text{Cl}^-/\text{K}^+/\text{Na}^+$ symporter in the thick ascending limb. Since water follows ions, and ion reabsorption is impaired, increased diuresis (urination) occurs. Volume loss causes upregulation of RAAS and distal secretion

	of protons under the control of aldosterone, leading to metabolic alkalosis. This is referred to as “contraction alkalosis,” where volume contraction induced by diuretics promotes metabolic alkalosis. In addition, additional distal secretion of K^+ occurs. - Thiazide diuretics will also follow this arrow pattern. They block the Na^+/Cl^- symporter on the apical membrane of the early-DCT, leading to volume depletion and $\uparrow$ RAAS, thereby $\uparrow$ H^+ and K^+ secretion distally. - CO_2 is acidic and retained because it compensates for our metabolic alkalosis. - It should be noted that these arrows are the same as for vomiting.
Diarrhea / Laxatives	- $\downarrow$ Serum K^+, $\uparrow$ Cl^-, $\downarrow$ HCO_3^-, $\downarrow$ pH, $\downarrow$ CO_2, $\leftrightarrow$ anion gap. - Stool is rich in potassium and bicarb. Diarrhea and laxative abuse can cause a normal anion-gap metabolic acidosis. Since CO_2 is acidic, we blow this off to compensate. - Serum chloride is high due to cellular shifts. - Anion gap is normal because diarrhea is not part of MUDPILES.
Shock	- Whether it's septic, hypovolemic, cardiogenic, etc., USMLE wants you to know that we often get metabolic acidosis due to lactic acidosis. - If we aren't perfusing our tissues as well as we're supposed to, then anaerobic respiration increases, leading to lactic acidosis. - The NBME might say a patient had an MI, and you say, “why the fuck is the bicarb 20?” It's just due to lactic acidosis because we have cardiogenic shock. - If pH is low in pulmonary embolism, this is due to obstructive shock, where lactic acidosis has occurred. But $>6/7$ Qs, PE will just be acute respiratory alkalosis.

Endocrine pharm for USMLE (keeping things HY)	
Metformin	- MOA = “$\uparrow$ Glycolysis + $\downarrow$ gluconeogenesis.” - Only used for type II DM. - Causes lactic acidosis. - Discontinue if bicarb is low (means lactic acidosis). - Discontinue (or don't commence) if creatinine is 1.5 or greater, as reduced renal function $\uparrow$ risk of lactic acidosis. - Choose insulin over metformin for regulation of glycemic control in pregnancy / intrapartum, as it allows for optimal fine-tuning / control of glucose. - USMLE can give you patient on metformin who had CT with contrast + gets high creatinine as a result + they ask you what would have prevented the increase in creatinine prior to the CT $\rightarrow$ answer = “infusion of 0.9% saline” or “adequate intravenous hydration”; the wrong answer is “discontinuation of metformin.” This one is a little tricky, where you need to know that adequate hydration saline hydration is the #1 way to prevent contrast nephropathy. There is $\uparrow$ risk of lactic acidosis due to metformin in patients who have renal insufficiency, but metformin itself doesn't cause renal insufficiency. In other words, $\uparrow$ creatinine is a reason to stop metformin, but $\uparrow$ creatinine doesn't result from metformin itself.
Insulin	- USMLE doesn't give a fuck that you know about all of the different insulin types and their durations of action. - What USMLE might do on 2CK is give you a patient with diabetes who wakes up with severe hyperglycemia + ask you what needs to be done. The answer = decrease, not increase (holy shit), dose of evening intermediate-acting insulin. This is called Somogyi phenomenon, where it appears as though because morning glucose is elevated, this means evening/night insulin is insufficient. But it's actually the opposite, where the evening/night dose is too high, which pushes the patient into overnight hypoglycemia, causing a compensatory sympathetic response that results in a robust rebound hyperglycemia by morning. - The other Q they will give is a patient with type II DM who is already on metformin + glyburide who has uncontrolled HbA1c at 12% + give you either low bicarb or high creatinine $\rightarrow$ they ask what needs to be done $\rightarrow$ answer = “discontinue metformin and glyburide and commence intermediate-acting insulin.”

	I've made YouTube clips on this Q and have talked about it many times, as it gets asked in our Telegram group probably once a month. The point is that 1) if a patient has low bicarb while on metformin, this means lactic acidosis due to metformin, so you must choose an answer choice where metformin is discontinued; and 2) a patient who has a creatinine of 1.5 or greater needs to be off metformin (I've seen max 1.4 on NBME where metformin is acceptable), since renal insufficiency is a risk factor for lactic acidosis due to metformin.
Sulfonylureas	- Glyburide, glipizide, glibenclamide. - Insulin secretagogues – i.e., ↑ insulin secretion by blocking the ATP-gated K⁺ channel on β-islet cells without the need for ATP. - Glyburide and glipizide tend to be the ones that shows up repeatedly on NBME. I don't think I've ever seen other sulfonylurea on NBME for that matter. - Glibenclamide was known for causing severe hypoglycemia. You could be aware of this in theory.
Meglitinides	- Repaglinide. - Similar to sulfonylureas, these ↑ insulin secretion by blocking the ATP-gated K⁺ channel on β-islet cells without the need for ATP.
GLP-1 analogues	- Liraglutide, semaglutide (Ozempic). - Glucagon-like peptide-1 stimulates insulin release.
DPP-4 inhibitors	- Sitagliptin. - DPP-4 is an enzyme that breaks down GLP-1, so if we inhibit DPP-4, GLP-1 is ↑. - Therefore, ↑ insulin release from the pancreas.
Thiazolidinediones	- Pioglitazone. - ↑ insulin sensitivity at tissues by stimulating PPAR-γ. - Can cause liver toxicity.
α-glucosidase inhibitors	- Acarbose, miglitol - ↓ Glucose absorption in small bowel. - Can cause diarrhea.
SGLT-2 inhibitors	- Dapagliflozin. - ↓ Glucose reabsorption in the PCT of the kidney. - Can ↑ risk of candidal infections.
Amylin analogues	- Pramlintide. - Amylin is a hormone that is co-secreted with insulin from β-cells that suppresses glucagon secretion, slows gastric emptying, and promotes satiety, thereby preventing a rapid increase in postprandial blood glucose.
PTU/methimazole	- Propylthiouracil (PTU) and methimazole are both used for Graves. - Both inhibit thyroperoxidase; PTU has an additional MOA of inhibiting 5-deiodonase. - Both PTU and methimazole can cause neutropenia/agranulocytosis. You need to know this often presents as mouth ulcers on USMLE. For example, they'll say patient was recently treated in hospital for thyroid storm + now has mouth ulcers and fever → answer = "drug adverse effect," or "drug-induced neutropenia." - Propranolol is used for tachycardia in hyperthyroidism (beta-blockade inhibits 5-deiodonase). Steroids are used for exophthalmos. - Methimazole is teratogenic in first-trimester.
Demeclocycline	- Shows up as Tx for SIADH on NBME. - Tetracycline antibiotic that causes nephrogenic DI as an adverse effect, but this is actually a "good thing" if we want to treat SIADH (i.e., let's just treat one problem by causing another problem, and they somehow cancel out).
Desmopressin	- Vasopressin (ADH) analogue used as treatment for central DI. - Also used after fluid restriction in DI cases for differentiating central from nephrogenic.
Octreotide	- Somatostatin analogue used for acromegaly and esophageal varices.
Cinacalcet	- Increases sensitivity of calcium-sensing receptors on parathyroid glands. - Helps to suppress PTH in secondary hyperparathyroidism.

If you've read the above, you've finished this PDF for all intents and purposes for USMLE. Now you can proceed to my HY Arrows PDF.

I'm adding the original bullet points from my old HY Endocrine PDF below because some whiny med students said they preferred this style instead. Hard to appease everyone.

- Na/K role of aldosterone → upregulates Na/K-ATPase pump on basolateral membrane of cortical collecting duct + late-DCT → reabsorbs 3Na into blood for secretion of every 2K into tubular cell (goes to urine)
- Acid-base role of aldosterone → causes direct proton secretion at apical membrane of cortical collecting duct
- How does aldosterone relate to ENac? → increased reabsorption of Na at basolateral membrane means decreased intracellular Na → favorable high-low gradient of Na from urine into tubular cell → ENac upregulates on apical membrane to increase movement of Na into tubular cell
- Acid-base / biochemical disturbance in Addison → low Na, high K, low bicarb, low pH (metabolic acidosis)
- Fatigued patient + hyperpigmentation + low BP → Addison disease (primary hypoadrenalism; low aldosterone + low cortisol)
- Infective cause of Addison → Waterhouse-Friderichsen syndrome (WFS)
- What is WFS → hemorrhagic (not ischemic) necrosis of adrenal glands
- Cause of WFS → meningococcal septicemia (non-blanching rash)
- Low BP in WFS + fluids are given; next best step? → give **dexamethasone** to compensate for low cortisol
- If WFS is hemorrhagic necrosis, what HY scenario contrasts → Sheehan syndrome is ischemic necrosis of anterior pituitary following labor
- Cause of Sheehan → anterior pituitary doubles in size during pregnancy to increase prolactin secretion; blood loss during labor → ischemic infarction
- USMLE arrow Q for Sheehan → down ACTH, down TSH, down prolactin, up aldosterone (NBME exam)
- How do you Dx Addison disease → ACTH stimulation test (if cortisol doesn't go up appreciably, Dx confirmed)

- Weird hematologic finding in Addison → **eosinophilia** (don't go chasing ova, stool, parasites)
- Why fatigue in Addison → low cortisol causes chronic fatigue syndrome
- Why hyperpigmentation in Addison → low cortisol, so less negative feedback at hypothalamus + anterior pituitary → increase in CRH → increase in POMC (precursor to ACTH and alpha-MSH)
- Tx for Addison → fludrocortisone
- Important point about fludrocortisone → corticosteroid with high mineralocorticoid effect (acts like aldosterone) → can be used to overcome low aldosterone + low cortisol caused by Addison
- Important point about prednisone + hydrocortisone + dexamethasone → glucocorticoids with very low mineralocorticoid effect (do not act like aldosterone) → hydrocortisone classically used to treat secondary hypoadrenalism (only decreased cortisol)
- Cause of Addison → autoimmune, typically part of polyglandular syndrome
- Cause of secondary hypoadrenalism → decreased ACTH production → usually due to impingement by pituitary tumor (e.g, prolactinoma); can also be caused by Sheehan syndrome
- ACTH levels in Addison vs secondary hypoadrenalism → high in Addison; low in secondary
- Renin levels in Addison? → high
- Potassium levels in secondary hypoadrenalism? → normal because aldosterone intact through RAAS; cause is merely decreased ACTH
- Primary hyperaldosteronism → Conn syndrome (tumor) or diffuse hyperplasia (autoimmune)
- Renin level in Conn syndrome → low
- High BP + high renin/aldosterone ratio in young woman → fibromuscular dysplasia (FMD)
- Dx of fibromuscular dysplasia? → MR angiography of renal arteries
- High BP + high renin/aldosterone ratio in older patient with cardiovascular disease → renal artery stenosis
- Cause of renal artery stenosis → atherosclerosis
- Patient with high BP + given ACEi + now creatinine increases; Dx? → renal artery stenosis or FMD
- Acid-base / biochemical disturbance in RAS or FMD → high Na, low K, high bicarb, high pH (metabolic alkalosis)

- How does RAAS work? → Low blood volume → JGC secrete renin → renin cleaves angiotensinogen (produced by the liver) in the plasma into angiotensin I → AT-I goes to lungs and is cleaved by ACE into AT-II → AT-II goes to zona glomerulosa of adrenal cortex → upregulates aldosterone synthase → converts 11-deoxycorticosterone into aldosterone
- Important functions of AT-II unrelated to aldosterone? → potent vasoconstrictor on systemic arterioles (increases afterload) + renal efferent arterioles (increases filtration fraction [GFR/RPF] in setting of low blood volume, meaning that GFR is maintained despite low renal plasma flow) + increases PCT reabsorption of Na
- Why is AT-II-mediated PCT reabsorption of Na important? → water follows Na; mechanism to increase fluid retention in setting of low blood volume
- Effect of giving ACEi on efferent arteriolar diameter → increases diameter
- ACEi (e.g., enalapril) effect on RAAS → increases renin, increases AT-I, decreases AT-II, decreases aldosterone
- ARB (e.g., valsartan) effect on RAAS → increases renin, increases AT-I, increases AT-II, decreases aldosterone
- Spironolactone effect on RAAS → increases renin, increases AT-I, increases AT-II, increases aldosterone
- Eplerenone vs spironolactone → eplerenone way less likely to cause gynecomastia
- If suspected Dx of Conn syndrome → CT abdomen
- If CT abdomen negative (hyperplasia not tumor) → adrenal venous sampling to confirm
- What is pheochromocytoma → catecholamine-secreting tumor of adrenal medulla
- Paroxysmal headaches/palpitations (high BP) + normal electrolytes → pheochromocytoma (PCC)
- Paroxysmal headaches/palpitations (high BP) + high glucose → PCC (catecholamines cause liver to make glucose)
- Tx for PCC → phenoxybenzamine first (irreversible alpha-1 blocker); never beta-blocker first
- Why phenoxybenzamine first to treat PCC → if you give beta-blocker first, you get “unopposed alpha,” meaning all of the NE + E (catecholamines) floating around bind to alpha-1, causing massive arteriolar vasoconstriction, shooting BP through the roof

- Dx of PCC → urinary or serum metanephrines
- Who gets PCC → NF1, MEN2A/2B (ret gene)
- PCC adults or kids? → adults
- PCC-like presentation in kids? → neuroblastoma (n-myc gene)
- Dancing eyes + high BP + midline mass in kid → neuroblastoma
- Dancing eyes? Wtf? → opsoclonus-myoclonus syndrome (seen in neuroblastoma; yes, weird)
- Dx of neuroblastoma → abdominal ultrasound
- Kid with painless flank mass that doesn't cross midline → Wilms tumor
- Dx of Wilms tumor → renal ultrasound
- Neonatal boy with midline suprapubic mass → posterior urethral valves
- Dx of posterior urethral valves → voiding cystourethrogram
- Cushing syndrome vs Cushing disease → syndrome = what you look like + refers to any cause of Cushingoid appearance; Cushing disease only = anterior pituitary ACTH-secreting tumor; in other words, Cushing disease is a cause of Cushing syndrome
- Most common cause of Cushing syndrome → exogenous glucocorticoids (e.g., prednisone)
- Pt not on exogenous steroids + Cushingoid; most common cause? → Cushing disease most common cause of endogenous Cushing syndrome
- Main causes of Cushing syndrome → exogenous steroids (most common overall), Cushing disease (most common endogenous), small cell bronchogenic carcinoma (ectopic ACTH), cortisol-secreting tumor (or diffuse hyperplasia) of zona fasciculata of adrenal cortex; CRH tumor rare as fuck
- Patient with chronic disease (i.e., IBD, SLE, RA) + Cushingoid; what are the ACTH + cortisol levels → need to know this means patient is taking prednisone → **low ACTH + low cortisol** (prednisone is **NOT** the same thing as cortisol) → prednisone suppresses CRH and ACTH secretion at hypothalamus and anterior pituitary → decreased endogenous cortisol production
- Patient with Cushing disease; ACTH + cortisol levels? → high ACTH + high cortisol
- Smoker + Cushingoid → small cell bronchogenic carcinoma
- Smoker + Cushingoid; ACTH + cortisol levels? → high ACTH (ectopic) + high cortisol
- When to do dexamethasone suppression test → Cushingoid patient not on exogenous glucocorticoids and has high serum cortisol

- Low-dose dexamethasone suppression test → tells us yes or no, patient has pathologic cause of Cushing syndrome (i.e., Cushing disease, or SCC of lung, or cortisol-secreting tumor), but we can't establish the causation from this; if cortisol doesn't suppress → yes, patient has true Cushing syndrome (proceed to high-dose test); if cortisol suppresses → no, patient does not have Cushing syndrome (do not proceed to high-dose test)
- High-dose dex → only cause of Cushing syndrome that will suppress in response is Cushing disease
- Pt has no suppression to low-dose dex + suppresses to high-dose → Dx = Cushing disease (ACTH secreting tumor of anterior pituitary)
- Pt has no suppression to low- or high-dose dex → ACTH high? → Yes, answer = SCC of lung; No → answer = cortisol-secreting tumor (or diffuse hyperplasia) of adrenal cortex
- Cushingoid + low ACTH + low cortisol → exogenous steroids
- Cushingoid + low ACTH + high cortisol → cortisol-secreting tumor (or diffuse hyperplasia) of adrenal cortex
- Cushingoid + high ACTH + high cortisol → Cushing disease
- Most accurate test for Dx Cushing syndrome → 24-hour urine cortisol
- Why dex test not most accurate? → false-positives in e.g., depression, alcoholism
- Acanthosis nigricans + low K + hyperpigmentation → Cushing syndrome
- Why acanthosis nigricans → caused by insulin resistance (unrelated: also can be caused by visceral malignancies)
- Why low K in Cushing syndrome → chronic elevation of glucocorticoid effect at kidney can push out potassium similar to aldosterone
- Why hyperpigmentation → high ACTH secretion means POMC is high → high alpha-MSH as well
- Purple striae in obese patient → Cushing syndrome
- Why purple striae → glucocorticoids weaken collagen → micro bleeding into skin
- Why hypertension in Cushing syndrome → cortisol upregulates alpha-1 receptors on arterioles, thereby allowing NE and E to do their job
- Graph shows you two scenarios: 1) NE given alone, then BP increases a little; 2) NE + cortisol given together, then BP increases a lot; why the difference? → cortisol is **permissive** of the effects of

catecholamines (don't choose synergistic or additive); once again, cortisol merely allows NE and E to do their job; cortisol isn't directly increasing BP

- Why normally ratio of E to NE in the blood is 80/20? → NE draining venously out of the adrenal medulla passes through the adrenal cortex → cortisol upregulates PNMT (converts NE to E)
- What does low cortisol cause? → chronic fatigue syndrome (as mentioned earlier but super important)
- Function of PTH → increases blood Ca + decreases blood PO₄
- Function of PTH at bone → increases bone resorption
- Cell PTH binds to at bone → **osteoblast, not -clast**
- If PTH causes bone resorption, why the fuck would it bind to osteoblasts (which build bone) → because PTH causes osteoblasts to express RANK-Ligand on their cell surface → binds to RANK receptor on osteoclasts → bone resorption
- How do osteoclasts resorb bone → intracellular carbonic anhydrase II (CAH-II) → creates H₂CO₃ from H₂O and CO₂; then the H₂CO₃ → bicarb + proton → protons accumulate at bone-osteoclast interface → surface for resorption
- What is teriparatide → N-terminus PTH analogue that binds to osteoblasts, and then rather than causing bone resorption, actually stimulates bone growth (difference in mechanism not well elucidated) → used in advanced osteoporosis
- Avoid teriparatide in whom? → Paget disease of bone patients; Hx of osteosarcoma
- Three effects of PTH at the kidney? → **1)** it upregulates 1-alpha hydroxylase activity in the PCT (converts inactive 25-OH-D₃ into active 1,25-(OH)₂-D₃; **2)** decreases PO₄ reabsorption in PCT by downregulating three types of apical membrane brush-border sodium-phosphate symporters; **3)** increases Ca reabsorption in the late-DCT by upregulating the apical TRPV5 transporter → reabsorbs calcium.
- What does secretion in kidney terms mean → excretion through the tubular wall (excretion is umbrella term → filtration = excretion through Bowman capsule; secretion = through tubular wall)
- Primary hyperparathyroidism biochemical disturbance? → high Ca, low PO₄, high ALP, high PTH
- Why is ALP high → ALP reflects osteoblast activity; if PTH high, then ALP also high (but annoyingly, if PTH is low, ALP will be normal, not low)

- Who gets primary hyperPTH → usually parathyroid adenoma (e.g., 22 yr old girl with nodule)
- Who else gets primaryPTH → MEN1 + MEN2A (can be diffuse 4-gland hyperplasia in MEN patients)
- Fuck, what are the MEN syndromes again? → Relax. MEN1 = parathyroid, pituitary, pancreas; MEN2A = parathyroid, PCC, medullary thyroid carcinoma; MEN2B = PCC, medullary thyroid carcinoma, mucosal neuromas, and Marfanoid body habitus (“oid” means looks like but ain’t)
- Question says girl has high Ca + low PO₄ + nodule of left, superior parathyroid gland; what’s the embryo derivative? → 4th pharyngeal pouch
- 3rd pouch = thymus + two inferior parathyroids
- 4th pouch = two superior parathyroids
- USMLE classically likes DiGeorge syndrome for agenesis of 3rd + 4th pouches, but will see if you can distinguish by asking parathyroid adenoma Qs
- Other weird info I need to know for primaryPTH? → Yeah, firstly, urinary Ca is high, not low.
- Wtf, how is that possible if PTH reabsorbs Ca from urine → because serum Ca is high, so the net amount in the urine is still high (this is a HY arrow Q that everyone gets wrong); in other words, on the USMLE, in primaryPTH → serum Ca up; serum PO₄ down; serum ALP up; urinary Ca **up (oh wow)**
- Second weird factoid about primaryPTH → urinary cAMP is elevated (USMLE likes this for some magical reason)
- What is the biochemical disturbance in secondary hyperparathyroidism? → Ca and PO₄ opposite of primary → low Ca; high PO₄, high ALP + high PTH
- Who gets secondaryPTH → chronic renal failure
- Why does chronic renal failure cause secondaryPTH → inability to activate vitamin D₃ in PCT → decreased Ca absorption through small bowel → low serum Ca → stimulates PTH release (this mechanism is on 2CK NBME interestingly); kidney also simply cannot reabsorb Ca as well, further decreasing serum Ca and stimulating PTH release
- What about low Ca in acute renal failure; doesn’t it take a while for vitamin D effects to occur → if low Ca in acute, due to mere inability to reabsorb from urine
- Why high PO₄ in renal disease → kidney can’t filter it out; even though there’s less activated D₃ and PO₄ absorption through small bowel is also decreased, the inability of the kidney to excrete it “wins”

- Biochemical disturbance of vitamin D deficiency → low Ca, low PO₄, normal/high PTH, normal/high ALP
- Function of vitamin D → increases Ca + PO₄ absorption in the bowel (don't worry about minimal role in bone)
- I don't get it though; if in chronic renal failure there's vitamin D deficiency due to less activation, why is there low Ca and high PO₄? Isn't vitamin D deficiency always low Ca and low PO₄? → chronic renal failure "wins" in terms of phosphate always → so although low vitamin D, there's still high PO₄.
- Any changes to the bones in a patient with chronic renal disease → renal osteodystrophy
- Hip fracture in renal failure + x-ray shows "pseudofractures" → vitamin D deficiency
- Osteitis fibrosa cystica → high PTH can cause "brown tumors" of bone + cholesterol accumulation in bone
- Kid with bowing of tibias + chondrochondritis + craniotables → rickets (vitamin D deficiency)
- Adult with pseudofractures → osteomalacia (vitamin D deficiency)
- Where does vitamin D deficiency start → stratum basale of skin (asked on USMLE)
- What is the sequence for production? → 7-dehydrocholesterol in skin goes to cholecalciferol via UV-B → then cholecalciferol goes to liver where it takes on a hydroxylation to make 25-hydroxy-D3 (calcidiol) → then this goes to the kidney where, via PTH activating 1-alpha-hydroxylase, it gets converted to active 1,25-(OH)₂-D3 → then this goes to small bowel to cause Ca + PO₄ absorption
- If person doesn't get sunlight, what can he/she not make → cholecalciferol (7-dehydro is wrong answer; this is on NBME)
- What about 7-dehydrocholesterol in relation to sunlight → we make this on our own, then UV-B converts it into cholecalciferol
- Alcoholic who eats plenty of dairy + gets sunlight + has vitamin D deficiency → answer = decreased hepatic hydroxylation
- Inject person with Ca; what happens to their vitamin D notably → answer = increased 24,25-(OH)₂-D3.
- Wtf is 24,25? → 25-OH-D3 is immediately converted to 24,25 as a storage form (think of them as the same)

- Inject with Ca → Ca binds to Ca-sensing receptor at parathyroid gland → negative feedback → causes reduction in PTH release → decreased 1-alpha-hydroxylase activity at kidney → buildup of 25 form → shunted to 24,25 form.
- Low calcium or potassium in a patient + he/she not responsive to supplementation → answer = hypomagnesemia → Mg needed for basal levels of PTH release (so low PTH → low Ca) + renal retention of K
- Alcoholic + low Ca or low K not responsive to supplementation → check serum magnesium
- Why low Mg in alcoholic → dietary deficiency
- Who gets high vitamin D (hypervitaminosis D) → **granulomatous disease (sarcoidosis)** → don't be the fool who says "girl tried to commit suicide by ODing on vitamin D pills?"
- Why high vitamin D in sarcoidosis → epithelioid (activated) macrophages secrete 1-alpha hydroxylase → activate vitamin D → increased small bowel absorption
- Ca and PTH levels in sarcoidosis → high Ca, **low PTH** (negative feedback)
- Sarcoidosis, any weird fact they ask about Ca? → Answer = "decreased Ca in feces" (makes sense)
- Biochemical disturbance in high vit D → high Ca, normal or high PO₄, normal or low PTH, normal or low ALP (USMLE will only give you one correct answer, don't worry, but I write the possibilities here so you don't get a Q where you see normal PTH or PO₄ and are wondering wtf, but Ca always up)
- Sarcoidosis other weird info → increased serum ACE (correct, angiotensin-converting enzyme; weird)
- Sarcoidosis, what happens first, high urinary Ca or high serum Ca → high urinary Ca (kidney will compensate before serum levels go up)
- 20s-30s African American woman with dry cough; CXR shows nodularity; Dx? → sarcoidosis; nodularity just means "bilateral lymphadenopathy"
- 20s-30s African American woman with dry cough; CXR shows nodularity; Dx? → "noncaseating granulomas" (sarcoidosis)
- 20s-30s African American woman with dry cough; CXR is normal Dx? → asthma (1/3 of asthma patients only have dry cough; this is called cough-variant asthma)
- 20s-30s African American woman with dry cough; CXR is normal Dx? → "increased activation of mast cells" (asthma)

- Mutation in calcium-sensing receptor on parathyroid gland → loss of negative feedback → PTH goes up, serum Ca up
- Mutation in calcium-sensing receptor on parathyroid gland → cause of familial hypocalciuric hypercalcemia (FHH)
- High PTH + high serum Ca + low urinary Ca → FHH
- High PTH + high serum Ca + high urinary Ca → primary PTH
- Mechanism for low urinary Ca in FHH? → not fully elucidated, but literature suggests increased tubular reabsorption, possibly paracellular, as serum Mg in FHH elevated relative to primary PTH
- What is cinacalcet → calcimimetic → activates Ca-sensing receptor at PTH gland
- When to use cinacalcet → tertiary hyperPTH
- What is tertiary hyperparathyroidism? → classically seen in renal transplant patients; autonomous four-gland hypersecretion of PTH following long-term secondary hyperparathyroidism and hypocalcemia → parathyroid chief cell hyperplasia that does not resolve with renal transplant → result is high Ca, variable PO₄ (high if still renal impaired), high ALP, high PTH
- What is pseudohypoparathyroidism? → insensitivity to PTH → high PTH but low Ca + high PO₄
- Anything special about pseudohypoparathyroidism? → **Yes. This is one of the highest yield yet underemphasized conditions on the USMLE Step 1. In other words, my students get these Qs on the exam regularly but the resources don't emphasize the different types of this condition**
- What is presentation of pseudohypoparathyroidism? → need to know Types 1a, 1b, 2
- Type 1a = **Albright Hereditary Osteodystrophy** = is simply the name of the phenotype → AHO has shortened 4th + 5th metacarpals, short stature + intellectual disability + **urinary cAMP does not increase with exogenous PTH administration** (normally should go up in response to PTH)
- Type 1b → high PTH + low Ca + high PO₄ + **no AHO phenotype** (just biochemical disturbance)
- Type 2 → same as Type 1b **but urinary cAMP increases in response to exogenous PTH**
- Mike, this sounds pedantic though. The USMLE really asks about pseudohypoparathyroidism like that? → Yeah. They ask it as arrow questions. And I personally had two pseudohypoPTH Qs on my Step 1 + my students do.
- What is pseudopseudohypoparathyroidism? → AHO phenotype but no biochemical disturbance
- Graves disease parameters → low TSH, high T₃, high T₄, increased iodine uptake

- Mechanism of Graves → **activating** TSH-receptor antibody, called TSI – thyroid stimulating immunoglobulin)
- Tx for Graves → propylthiouracil (PTU) or methimazole to address thyroid derangement
- PTU and methimazole MOAs → both inhibit thyroperoxidase, but PTU also inhibits peripheral conversion by inhibiting 5-deiodinase → propylthiouracil is a longer word than methimazole, so it's the one that does both
- Tx for tachycardia in Graves → propranolol (because beta-blockade also decreases peripheral conversion of T4 to T3)
- Important point about Graves? → proptosis/exophthalmos + pretibial myxedema are specific; in other words, there are numerous causes of hyperthyroidism (e.g., toxic multinodular goiter, toxic adenoma, etc.), but only Graves will cause the eye findings
- Tx for ophthalmopathy in Graves → **steroids**
- Why do the eye findings occur in Graves? → glycosaminoglycan deposition in/around extra-ocular muscles
- High BP + fever + increased CK → thyroid storm (Tx with PTU + propranolol)
- What is the role of potassium iodide (KI) in hyperthyroid Tx? → shuts off gland production (Wolff-Chaikoff effect) → answer in person exposed to nuclear fallout or radioiodine vapors in laboratory
- Hashimoto parameters → high TSH, low T3, low T4, decreased iodine uptake
- Mechanism for Hashimoto → antibodies against thyroperoxidase + thyroglobulin; anti-microsomal
- Histo of Hashimoto → lymphocytic infiltrate (easy to remember bc the non-eponymous name for Hashimoto is chronic lymphocytic thyroiditis)
- Tx for Hashimoto = levothyroxine (synthetic T4)
- 50M + low mood + BMI 26 → Hashimoto
- Proximal muscle weakness in Hashimoto + increased serum CK → hypothyroid myopathy
- 45M + decreased ability to get up from chair unassisted + HR of 60 → Hashimoto
- 45M + high cholesterol + high hepatic AST + HR of 55 → Hashimoto (hypothyroidism can cause bradycardia, high cholesterol, and high AST [the latter is weird, correct])
- Thyroid cancer in Hashimoto → thyroid lymphoma (autoimmune diseases → increased risk of lymphoma)

- 22M + viral infection + very tender thyroid → subacute granulomatous thyroiditis (de Quervain)
- De Quervain parameters → triphasic → causes hyper-, then hypo-, then rebounds to euthyroid state
→ most important point is that **iodine uptake is DECREASED in de Quervain + drug-induced thyroiditis, even if patient is in a hyper-phase.**
- Tx for de Quervain thyroiditis? → answer = NSAIDs
- 22M + very tender thyroid + HR of 88 + tremulousness + heat intolerance → low TSH, high T3, high T4, **decreased iodine uptake** (in contrast to Graves, which is painless and uptake is high)
- 27F + gave birth to healthy boy 6 months ago following uncomplicated labor + no weight change or mood disturbance + on no meds + vitals WNL + dry skin + thyroid gland enlarged and non-tender + TSH high + T4 low; most likely explanation for these findings? → answer = “thyroiditis” → Dx = **postpartum thyroiditis** (a type of **silent thyroiditis**) → characterized by thyrotoxicosis followed by hypothyroidism (1/3 of women experience both phases; 1/3 experience just hyperthyroid phase; 1/3 only hypothyroid phase); affects 5-10% of women postpartum; hyperthyroid phase usually occurs 1-4 months postpartum; hypothyroid phase occurs about 4-8 months postpartum; thought to be caused by postpartum immunologic rebound (immune system normally suppressed during pregnancy); Dx w/ Hx + ordering serum TSH; increased risk of progression to Hashimoto; Tx w/ short course of propranolol if hyperthyroid; give short course of levothyroxine if hypothyroid
- Tx for subacute thyroiditis → aspirin first, not steroids; steroids may be used later
- Drugs causing thyroiditis → lithium + amiodarone
- Surreptitious thyrotoxicosis → self- injection of thyroxine → low TSH, high T3, high T4, small thyroid gland with decreased uptake
- Injection of triiodothyronine (T3) → TSH will go down, T3 goes up (clearly), **T4 does not go up because T3 isn't converted to T4; only T4 is converted to T3**
- Injection of thyroxine → TSH will go down (negative feedback), T4 goes up (clearly), T3 goes up (due to peripheral conversion), reverse T3 also goes up
- What is reverse T3? → an inactive form of T3; T4 is converted peripherally into T3 (active) and reverse T3 (inactive)
- Anything else I need to know about reverse T3? → it's increased in euthyroid sick syndrome → times of stress/surgery/illness → cortisol increases → cortisol decreases conversion of T4 to active T3, so

more T4 is converted to reverse T3 → parameters in euthyroid sick syndrome: normal TSH, normal T4, low T3, high reverse T3

- What is subclinical hypothyroidism → high TSH but normal T3 + T4 (don't confuse with ESS)
- Subclinical hypothyroidism Tx → don't treat unless TSH >10 (normal is 0.5-5), Hashimoto Abs are present, or patient is pregnant → they ask this info on 2CK + Step 3
- Want to check thyroid function, what's the first thing to order → TSH
- Want to check thyroid function **in pregnancy**, what's the first thing to order → free-T4
- What is free-T4 → most thyroid hormone is protein-bound and inactive; free-T4 tells you definitively whether the patient has thyroid derangement or not
- Pregnancy and thyroid → estrogen causes increased thyroid-binding globulin production by the liver → mops of T4 → less free-T4 → less negative feedback at hypothalamus + anterior pituitary → TSH goes up transiently to compensate → more T4 made → free-T4 rebounds to normal but now total T4 is high → parameters you need to know for pregnancy: normal TSH + high total T4 + normal free-T4 + normal free T3 + high thyroid-binding globulin
- Hyperthyroidism in pregnancy → LH, FSH, TSH, hCG all share same alpha-subunit; their beta-subunits differ; some women have increases sensitivity of TSH receptor to alpha-subunit, so high hCG in early pregnancy can stimulate thyroid gland and cause transient hyperthyroidism
- Hashimoto in pregnancy → increase pregnant woman's dose of levothyroxine by 50%
- Graves in pregnancy → avoid methimazole in first trimester (teratogenic; causes aplasia cutis congenita) → give PTU in first-trimester → in second + third trimesters, switch from PTU to methimazole (methimazole no longer teratogenic later in pregnancy + PTU is heavily hepatotoxic)
- Pt being treated for Graves + mouth ulcers → agranulocytosis (neutropenia) caused by methimazole or PTU.
- Young child with normal free-T4 and low total T4 → thyroid-binding globulin deficiency (opposite of pregnancy)
- Young child + large belly + large tongue + hypotonia → cretinism (congenital hypothyroidism)
- Most common cause of cretinism → iodine deficiency
- Most common cause of hyperthyroidism in elderly → toxic multinodular goiter
- Evaluation of thyroid cancer, first step? → palpation of thyroid gland (on FM 2CK form as answer)

- If thyroid nodule present, then check TSH; if TSH normal or high → answer = ultrasound first, then FNA; if TSH low, do radioiodine uptake scan; thyroid cancer is cold, not hot, which is why no ultimate FNA with low TSH
- Struma ovarii → thyroid hormone secreting ovarian germ cell tumor
- Diabetic ketoacidosis parameters → low serum Na, high serum K (hyperkalemia), low total body K, low bicarb, low pH, low CO₂ (all exceedingly HY)
- Why low serum Na in DKA → osmotic effect of high glucose in blood → dilutional hyponatremia; in addition, there's believed to be a contributory renal mechanism related to increased absorption of water in the PCT due to increased SGLT2 reabsorption of glucose in PCT
- Why high serum K in DKA → three main reasons: 1) insulin normally drives K into cells, so if insulin isn't there, K is higher in blood; 2) less glucose driven into cells by GLUT4 (bc normally upregulated by insulin) means less ATP production → normally 1 ATP drives 2K into cell and 3Na out; so if less ATP-ase activation, less K enters cell → higher in blood; 3) potassium-proton exchange; if acidosis ensues, more H driven into cells means K moves out to balance charge → hyperkalemia
- Then what does low total body potassium mean → just to be clear, the patient is hyperkalemic (high K in the blood) yet has low K overall in the body → kidney senses high K in urine and therefore increases excretion of it (kaliuresis) → body is now losing K → but three above mechanisms leading to hyperkalemia continue unabated, so K stays high in serum even though body is now urinating it out.
- Why does the potassium stuff matter so much with DKA → because when you Tx DKA and start giving insulin (**fluids first btw**; giving insulin immediately is the wrong answer; give insulin after first administering a bolus of normal saline), K will now be driven into the cells, which will bring K down to normal in the serum, but bear in mind it was low in the cells → so now risk of normal in cells but low in blood → need to supplement K to patient when K falls below 5.2 (normal is 3.5-5 mEq/L). Stop all insulin if K falls below 3.3.
- Overweight, middle-age patient with polydipsia + polyuria → type II DM
- High in serum in type II DM → **insulin is high initially**; ketones absent (only ketones in DKA; DKA is type I only)
- Type II diabetic crisis? → hyperosmolar hyperglycemic non-ketotic syndrome (HHNS) → still give fluids first

- Glucose numbers in DKA vs HHNS → low-hundreds for DKA (i.e., 2-300s); can be 600-1000 for HHNS
- Acid-base disturbance in aspirin toxicity first 20 mins → resp. alkalosis (low CO₂, high pH, normal O₂, normal bicarb [too acute to change])
- Acid-base disturbance in aspirin toxicity after 20 mins → mixed metabolic acidosis-respiratory alkalosis (low CO₂, low pH, normal O₂, low bicarb)
- Tx for aspirin toxicity → bicarb (increased excretion through urinary alkalization)
- Acid-base disturbance in PE → resp. alkalosis (low CO₂, high pH, low O₂, normal bicarb [too acute to change])
- Acid-base disturbance in asthma → resp. alkalosis (low CO₂, high pH, low O₂, normal bicarb [too acute to change])
- Electrolyte abnormality in anorexia → hypokalemia
- Most common cause of death in anorexia → arrhythmia from hypokalemia
- Refeeding syndrome (even if normal BMI) → hypophosphatemia
- Amenorrhea in anorexia → low FSH + low estrogen (hypogonadotropic)
- Premature ovarian failure + Turner syndrome + menopause → high FSH (low inhibin) + low estrogen
- High BMI female + irregular menstrual cycles → anovulation
- Anovulation + hirsutism → PCOS
- Anovulation. Cause USMLE wants? → insulin resistance → causes abnormal GnRH pulsation
- Why hirsutism in anovulation → abnormal GnRH pulsation causes high LH/FSH ratio
- Why high LH/FSH ratio important in anovulation/PCOS → ovulation stimulated when follicle not ready → no ovulation (anovulation) → follicle retained as cyst
- What's LH do? → Stimulates theca interna cells (females) and Leydig cells (males) to make androgens
- What's FSH do? → Stimulates granulosa cells (females) and Sertoli cells (males) to make aromatase; also primes follicles
- Tx for PCOS → if high BMI, weight loss first always on USMLE
- Tx for PCOS if they ask for meds and/or weight loss already tried → OCPs (if not wanting pregnancy); clomiphene (if wanting pregnancy)

- PCOS increases risk of what → endometrial cancer (unopposed estrogen)
- Where does ADH (vasopressin) act → medullary collecting duct → causes aquaporin insertion on apical membrane → enables free H₂O reabsorption
- Where is ADH made? → supraoptic nucleus of hypothalamus → merely stored in posterior pituitary
- Where is oxytocin made → paraventricular nucleus of hypothalamus → merely stored in p. pituitary.
- Neurophysins I've heard about. What are those → Showed up in UWorld → carrier proteins needed to transport oxytocin + ADH from hypothalamus to posterior pituitary
- When does ADH go up → when serum sodium too high → brings sodium back down; ADH will also be secreted in response to lower blood volume, although aldosterone is major volume regulator; ADH is main tonicity regulator
- When considering SIADH vs diabetes insipidus (DI) vs psychogenic polydipsia (PP) → what's the next best step in evaluation/management → fluid restriction
- SIADH important causes → small cell bronchogenic carcinoma ectopic ACTH, or head trauma (can actually cause SIADH or DI)
- SIADH parameters → high urine osmolality (concentrated) + low serum sodium (normal is 135-145)
- Tx for SIADH → if small cell lung cancer, **chemotherapy** (HY to know you can't do surgery for small cell); if insufficient, give -vaptans (conivaptan, tolvaptan), which are ADH receptor antagonists, or **demeclocycline** (a tetracycline antibiotic that causes nephrogenic DI, but is a Tx for SIADH).
- Central diabetes insipidus → lack of production by hypothalamus or storage problem of posterior pituitary → low ADH + low urine osmolality (dilute urine) + high serum sodium (concentrated serum)
- Cause of CDI → idiopathic, local malignancy, or head trauma
- Nephrogenic DI → lack of sensitivity of kidney V₂ receptors to ADH → **high ADH**, low urine osmolality (dilute urine) + high serum sodium (concentrated serum)
- Cause of NDI → **lithium**, demeclocycline
- Patient with bipolar disorder + tremors + polyuria + polydipsia → answer = NDI
- Diabetes insipidus urine parameters relative to serum: PCT is isotonic (same; always unchanged), medullary collecting duct is hypotonic (dilute compared to serum), juxtaglomerular apparatus (JGA) is

hypotonic (always hypotonic no matter what the patient's condition bc value measured at the top of thick ascending loop of Henle after ions have been absorbed out of urine)

- SIADH urine parameters relative to serum: PCT is isotonic (same; always unchanged), medullary collecting duct is hypertonic (reabsorbing lots of free H₂O), JGA is hypotonic (as discussed above)
- Dude jumps into cold lake; what happens to central blood volume + atrial natriuretic peptide (ANP) + ADH levels? → CBV up (cold → sympathetic activation → alpha-1 agonism peripherally to decrease surface area of blood vessels to conserve heat → blood forced to core) + ANP up (if CBV up, then right atrial stretch up; ANP is body's natural diuretic → causes PCT to decrease Na reabsorption) + ADH down (baroreceptor at carotid sinus senses greater stretch → has a role not just on HR but also ADH release)
- Psychogenic polydipsia (PP)? → person drinks too much → low serum sodium + low urine osmolality
- Prolactin does what → milk production → acts through JAK/STAT tyrosine kinase
- Oxytocin does what → milk letdown/secretion
- Growth hormone in young → gigantism; in adults → acromegaly
- Important points about acromegaly → causes diabetes mellitus (GH causes insulin resistance), hypertension, carpal tunnel syndrome, arthritis, cardiomyopathy; and yes, prognathism (lantern jaw)
- Growth hormone acts directly at tissues? → USMLE wants you to know it causes liver to increase insulin-like growth factor 1 (IGF-1), which acts peripherally
- Which hormone counteracts GH → somatostatin → generally acts to shut off other hormone secretions, and antagonizes the effects of GH (can be Tx for acromegaly)
- What is pegvisomant? → GH receptor antagonist
- Congenital adrenal hyperplasia (CAH) → caused by 21, 11, or 17 hydroxylase deficiency in adrenal cortex (21 is 90-95% of cases) → autosomal recessive conditions
- 21 hydroxylase deficiency → severe lack of mineralocorticoid and glucocorticoid production by adrenal → low BP + high K + high DHEA-S (adrenal androgen)
- 11 hydroxylase deficiency → adrenal can still make 11-deoxycorticosterone in zona glomerulosa + 11-deoxycortisol in zona fasciculata → BP not low (sometimes high) + K not high; DHEA-S still high
- 17 hydroxylase deficiency → can't convert pregnenolone + progesterone to 17-OH forms → adrenal gland makes aldosterone in excess but not cortisol or DHEA-S

- What does ACTH do at adrenal gland → upregulates desmolase, which converts cholesterol into pregnenolone to start adrenal hormone synthesis
- What does angiotensin II do at the adrenal gland → upregulates aldosterone synthase, converting 11-deoxycorticosterone into aldosterone
- What is metyrapone testing? → 11-beta hydroxylase inhibitor → can be used in the diagnosis of adrenal insufficiency or Cushing → re the former, if you give metyrapone, cortisol should go down normally and ACTH + 11-deoxycorticosterone should go up; if ACTH goes up but 11-deoxycorticosterone doesn't → adrenal dysfunction (Addison); if ACTH doesn't go up, then it's secondary hypoadrenalism)



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